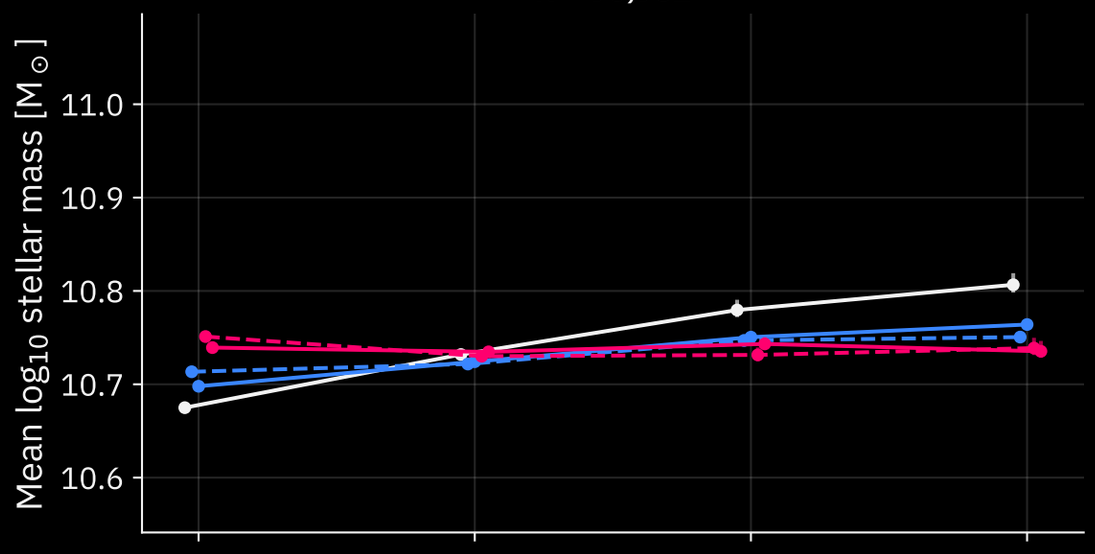
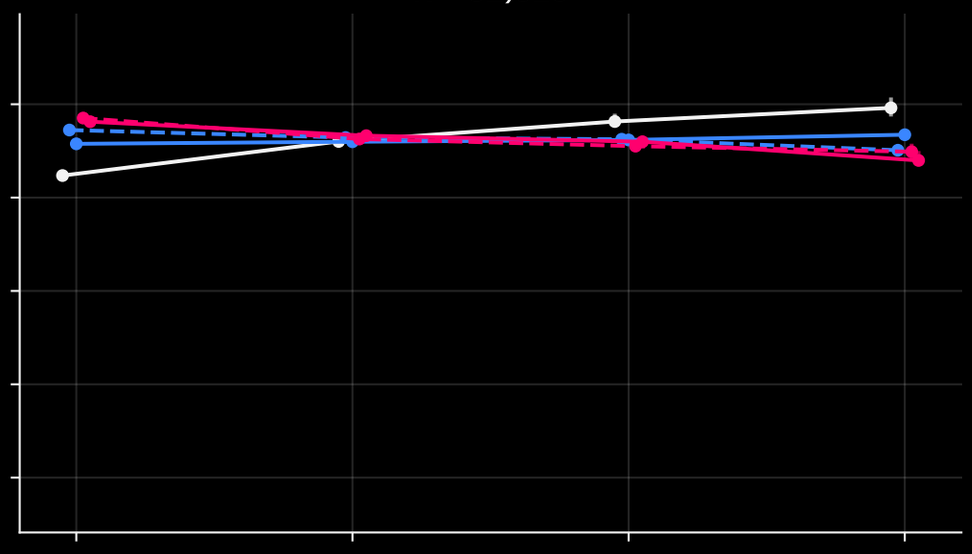
Actual TNG GAT versus Abacus P12-A · mass
soft_raw · Historical observed-feature normalization

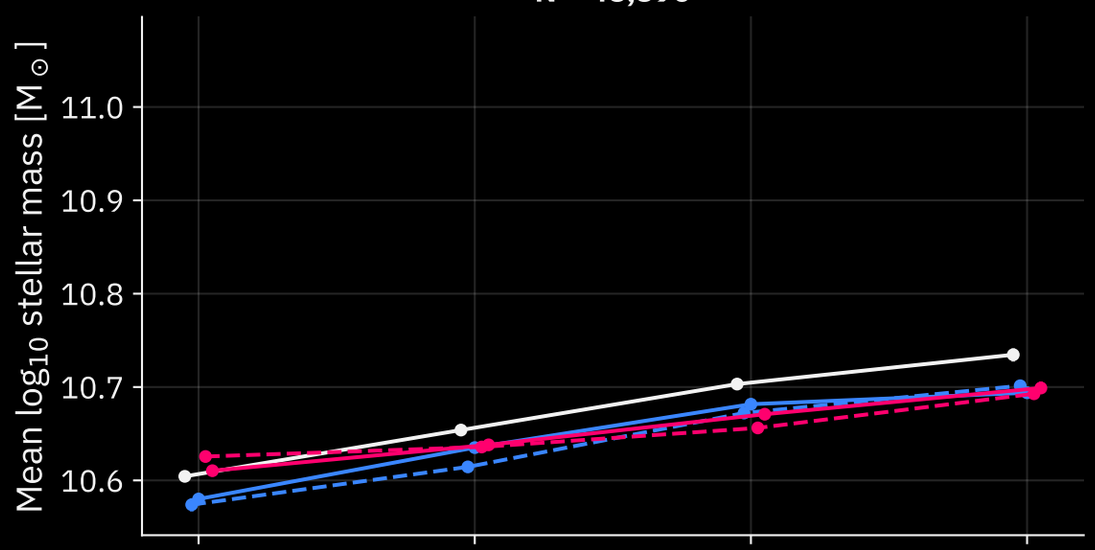
0.20 ≤ z < 0.30 · No completeness cut
N = 90,952



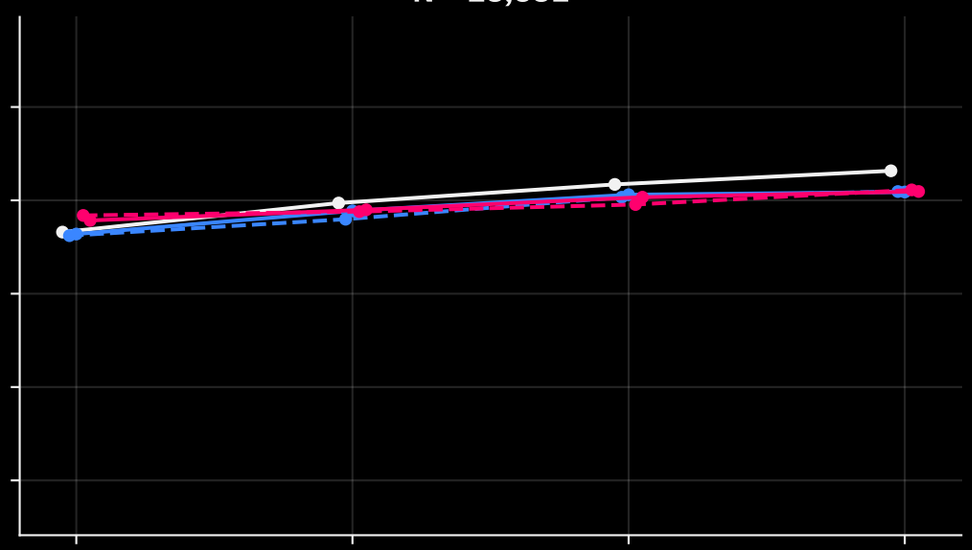
0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620



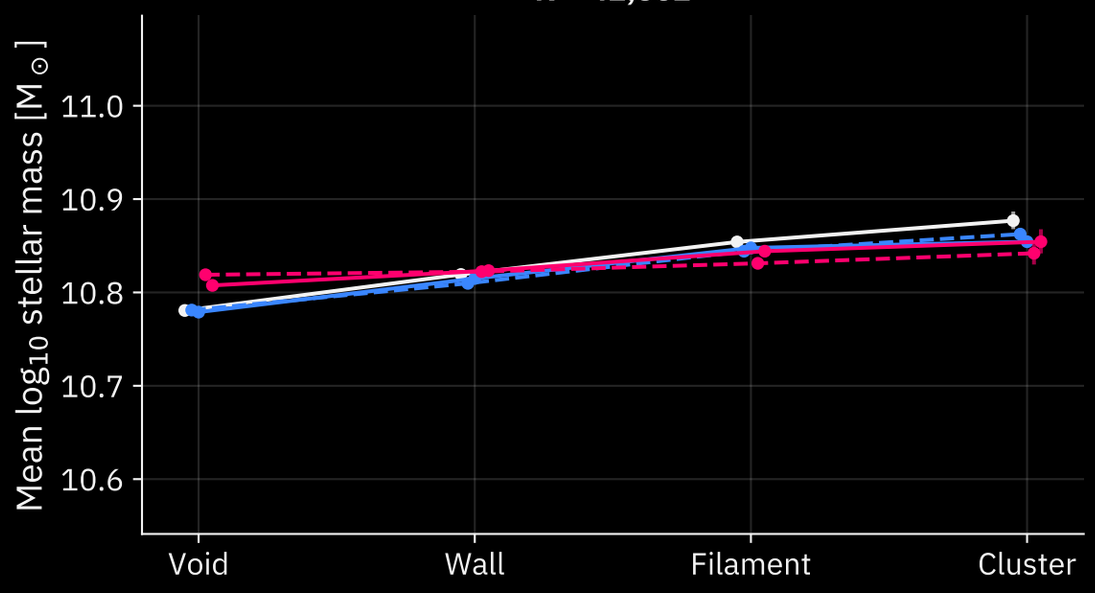
0.20 ≤ z < 0.25 · No completeness cut
N = 48,390



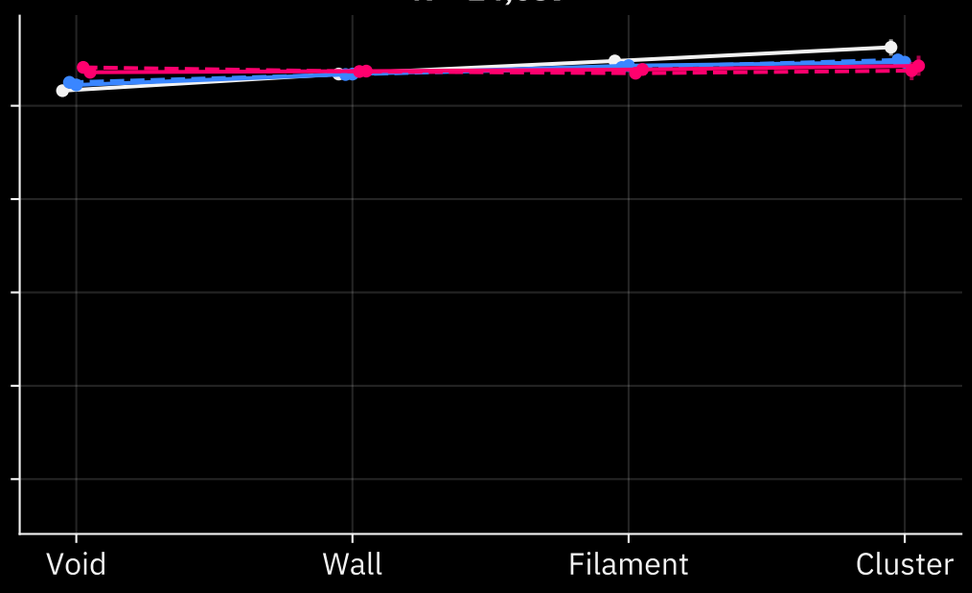
0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581



0.25 ≤ z < 0.30 · No completeness cut
N = 42,562



0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039



Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · sfr
soft_raw · Historical observed-feature normalization

0.20 ≤ z < 0.30 · No completeness cut
N = 90,952

0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620

0.20 ≤ z < 0.25 · No completeness cut
N = 48,390

0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581

0.25 ≤ z < 0.30 · No completeness cut
N = 42,562

0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039

Mean $\log_{10}$ SFR [$M_\odot$ yr^{-1}]

0.2
0.0
−0.2
−0.4
−0.6

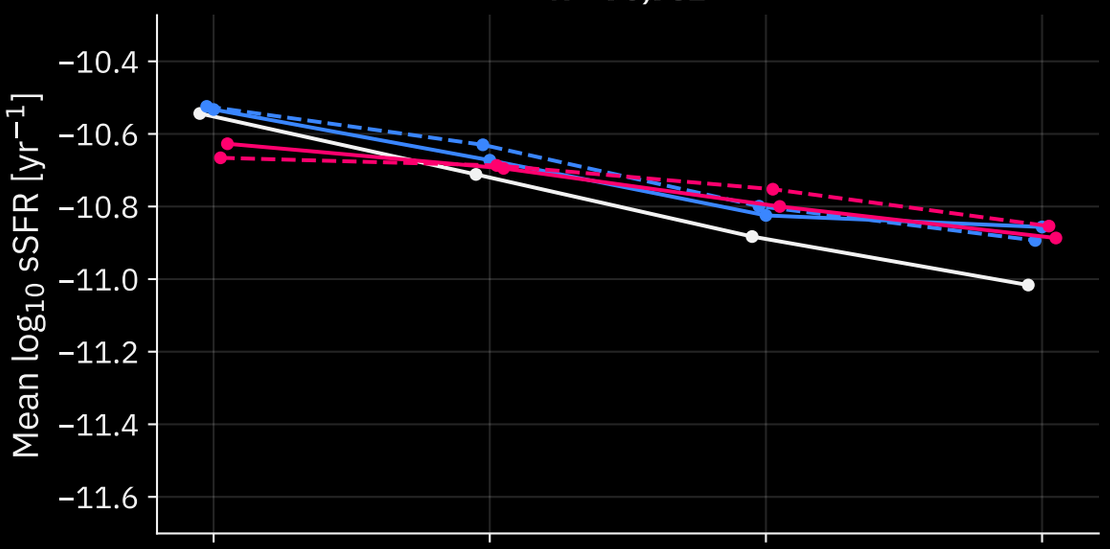
Void
Wall
Filament
Cluster

Abacus P12-A · R7
GAT · R2 · normal
GAT · R2 · thinned
GAT · R7 · normal
GAT · R7 · thinned

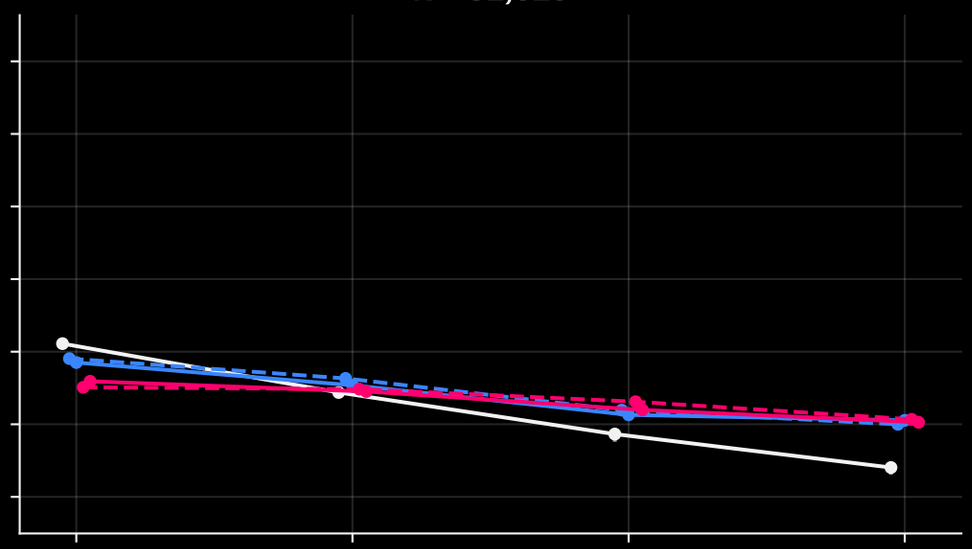
One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · ssfr
soft_raw · Historical observed-feature normalization

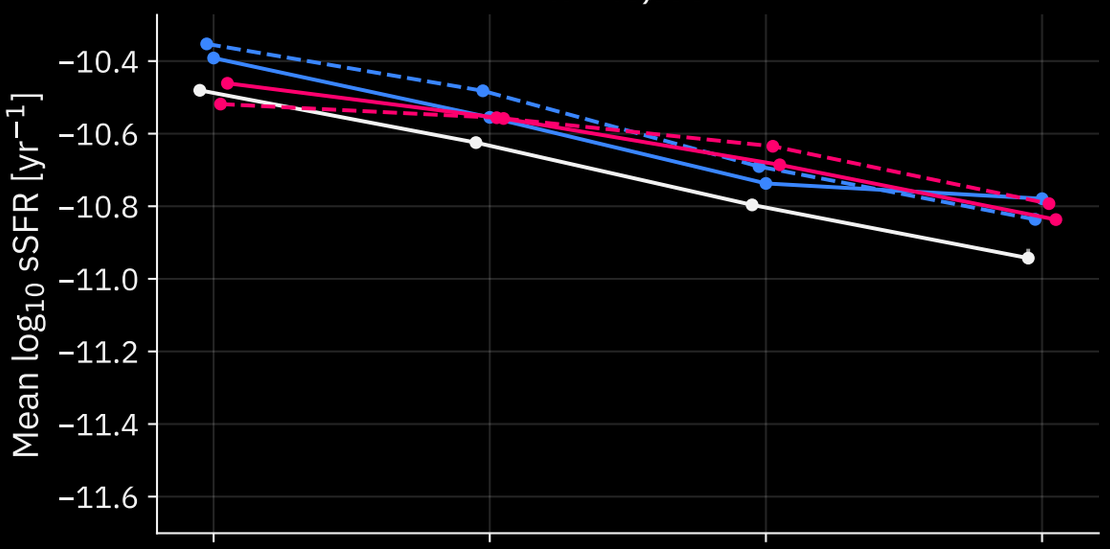
0.20 ≤ z < 0.30 · No completeness cut
N = 90,952



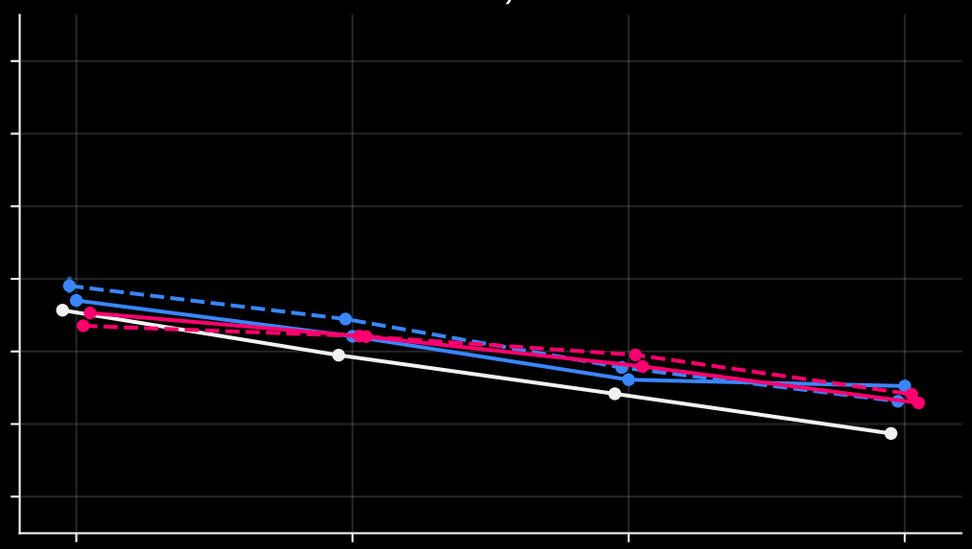
0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620



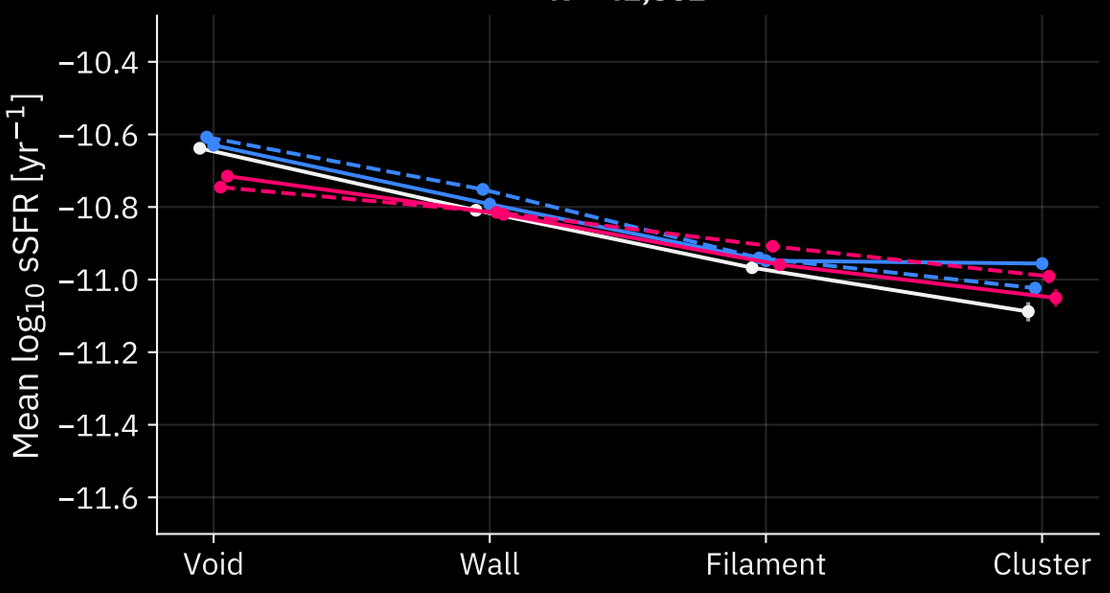
0.20 ≤ z < 0.25 · No completeness cut
N = 48,390



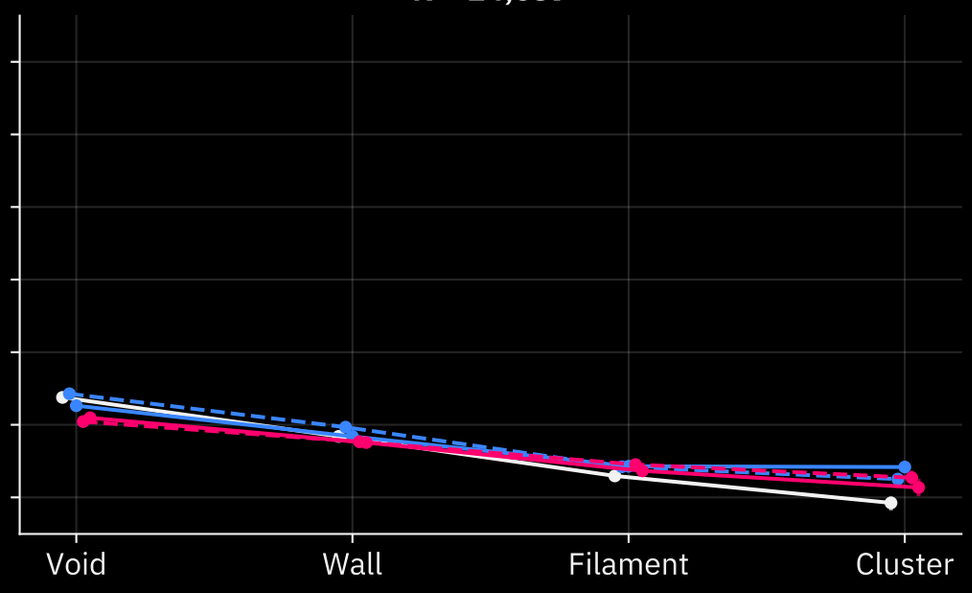
0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581



0.25 ≤ z < 0.30 · No completeness cut
N = 42,562



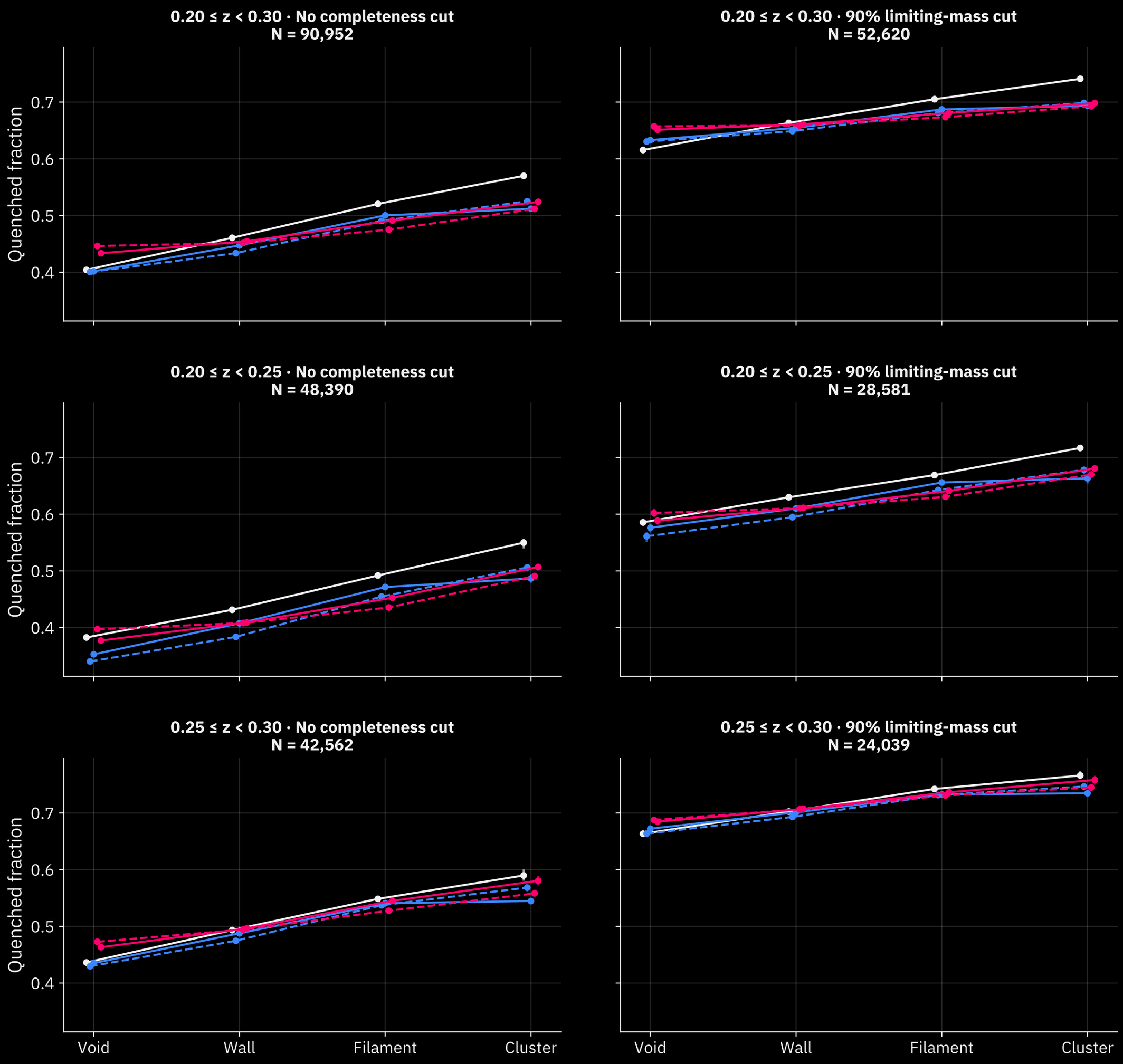
0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039



Abacus P12-A · R7
GAT · R2 · normal
GAT · R2 · thinned
GAT · R7 · normal
GAT · R7 · thinned

One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

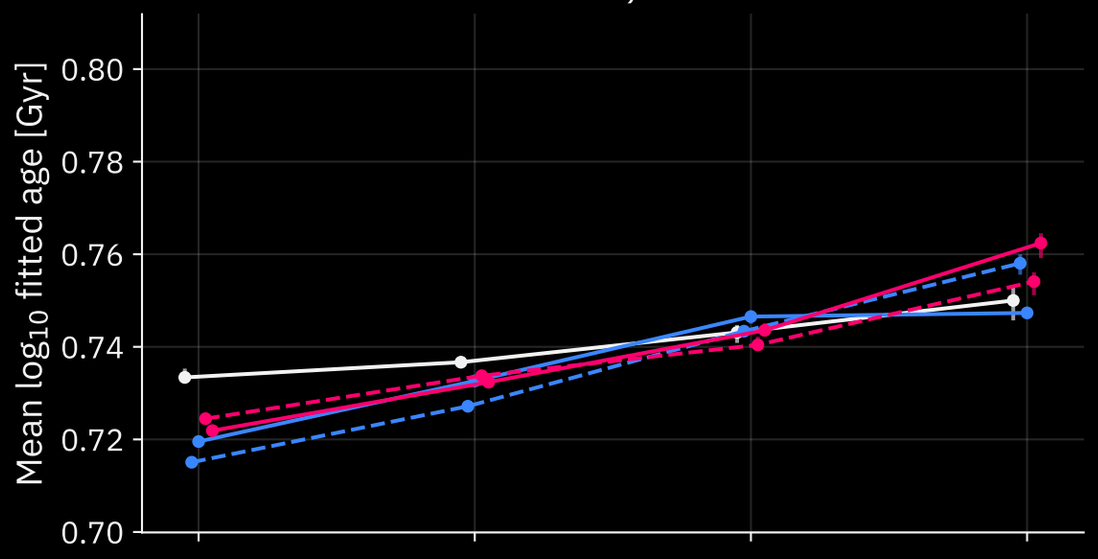
Actual TNG GAT versus Abacus P12-A · quenched
soft_raw · Historical observed-feature normalization



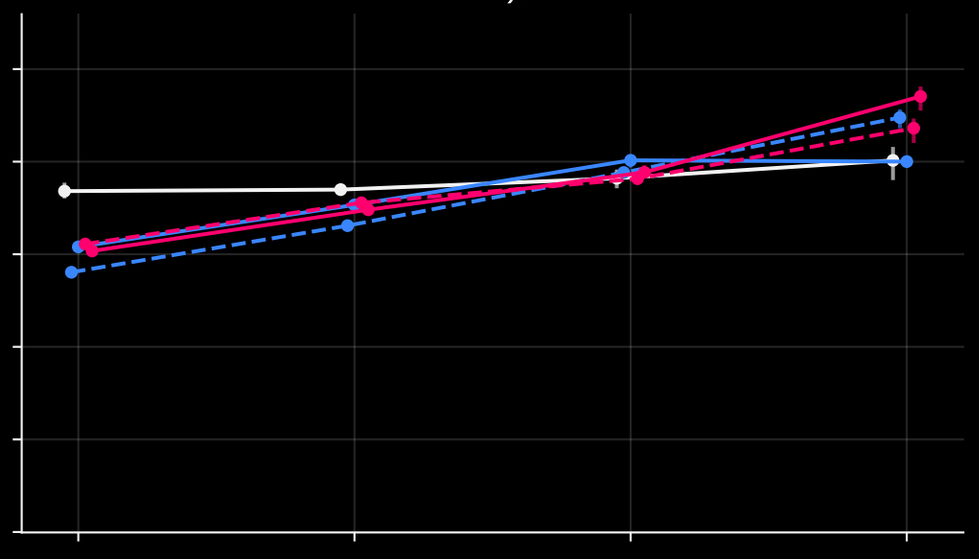
One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · age
soft_raw · Historical observed-feature normalization

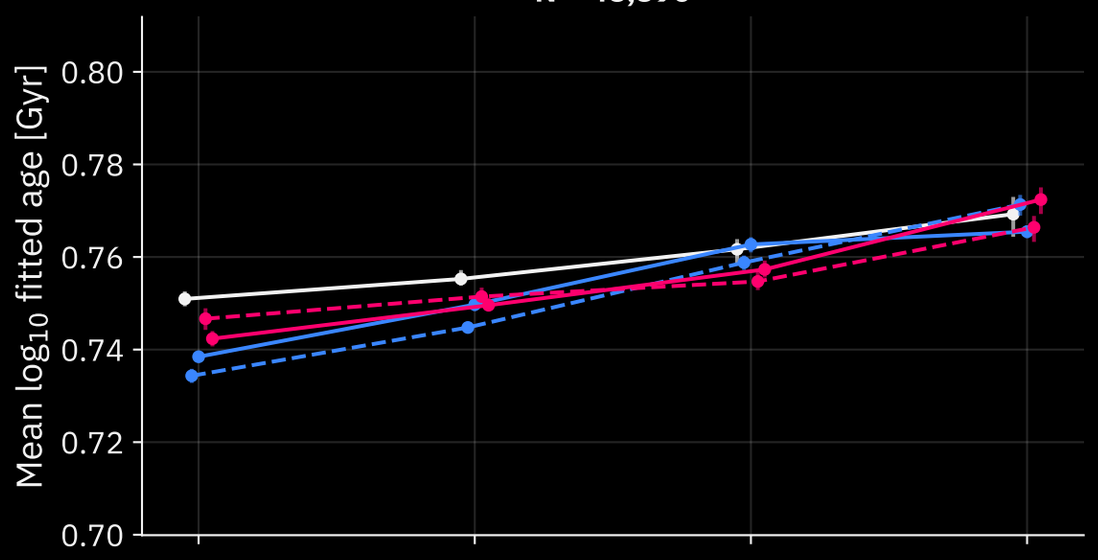
0.20 ≤ z < 0.30 · No completeness cut
N = 90,952



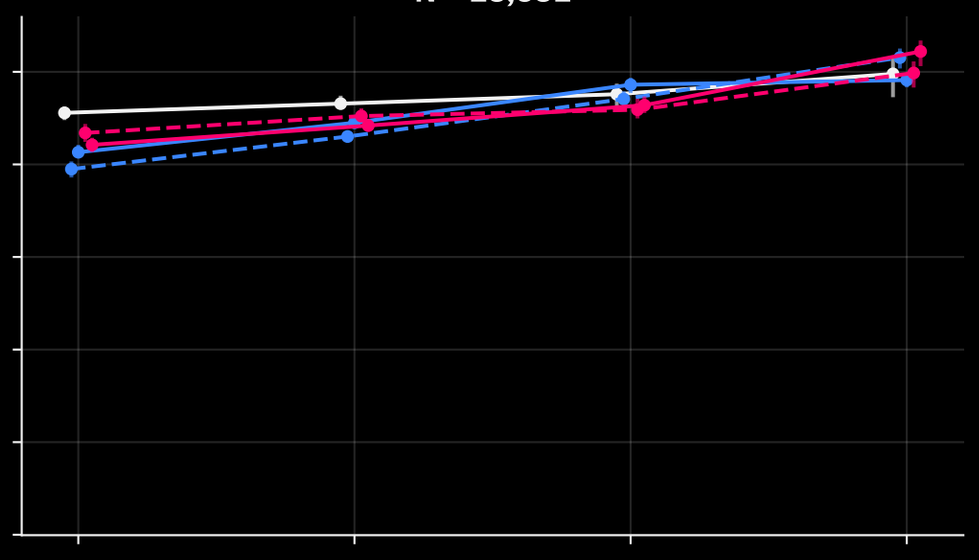
0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620



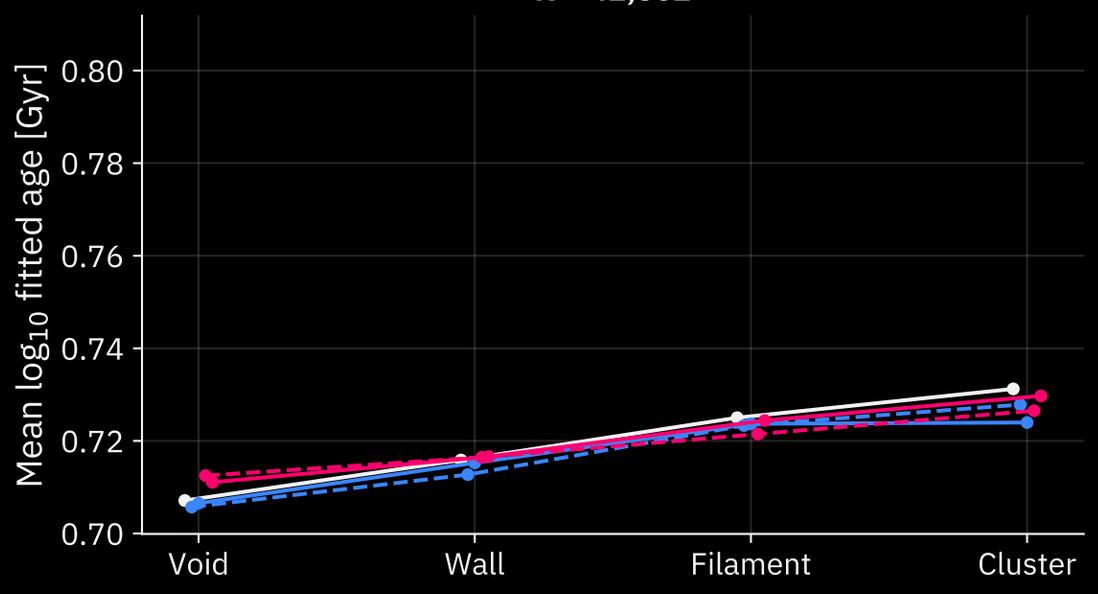
0.20 ≤ z < 0.25 · No completeness cut
N = 48,390



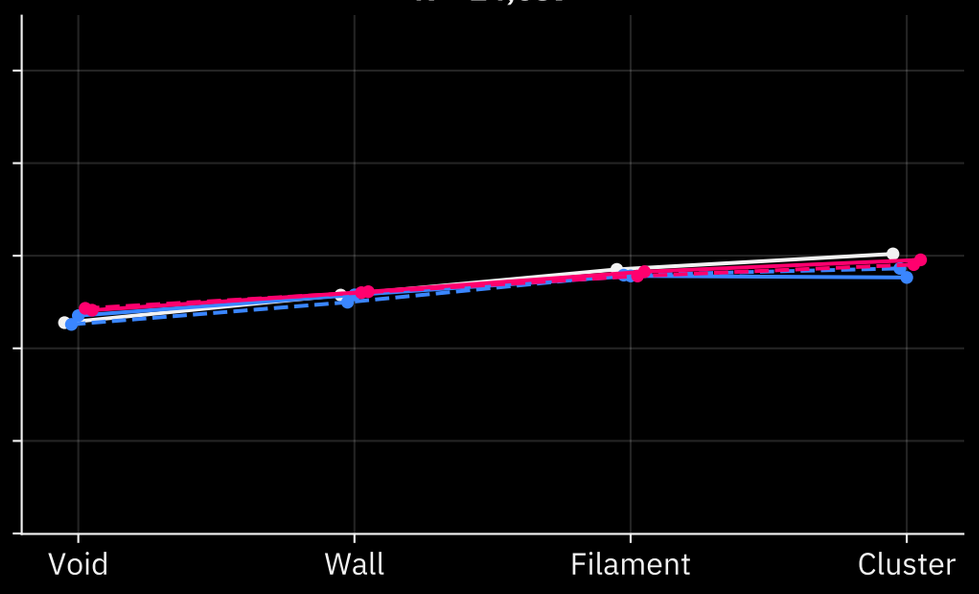
0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581



0.25 ≤ z < 0.30 · No completeness cut
N = 42,562



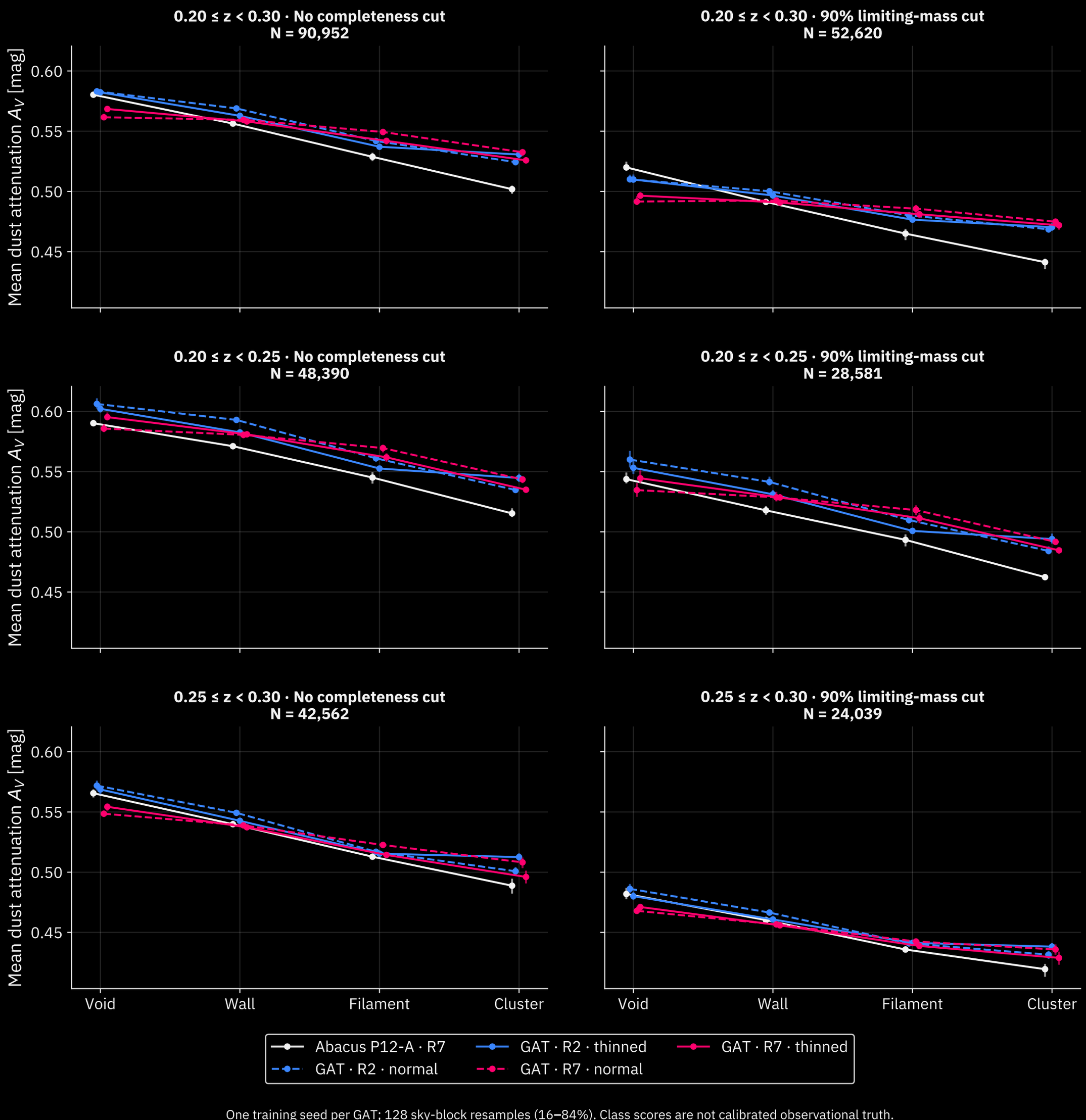
0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039



Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · av
soft_raw · Historical observed-feature normalization



One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · fagn
soft_raw · Historical observed-feature normalization

0.20 ≤ z < 0.30 · No completeness cut
N = 90,952

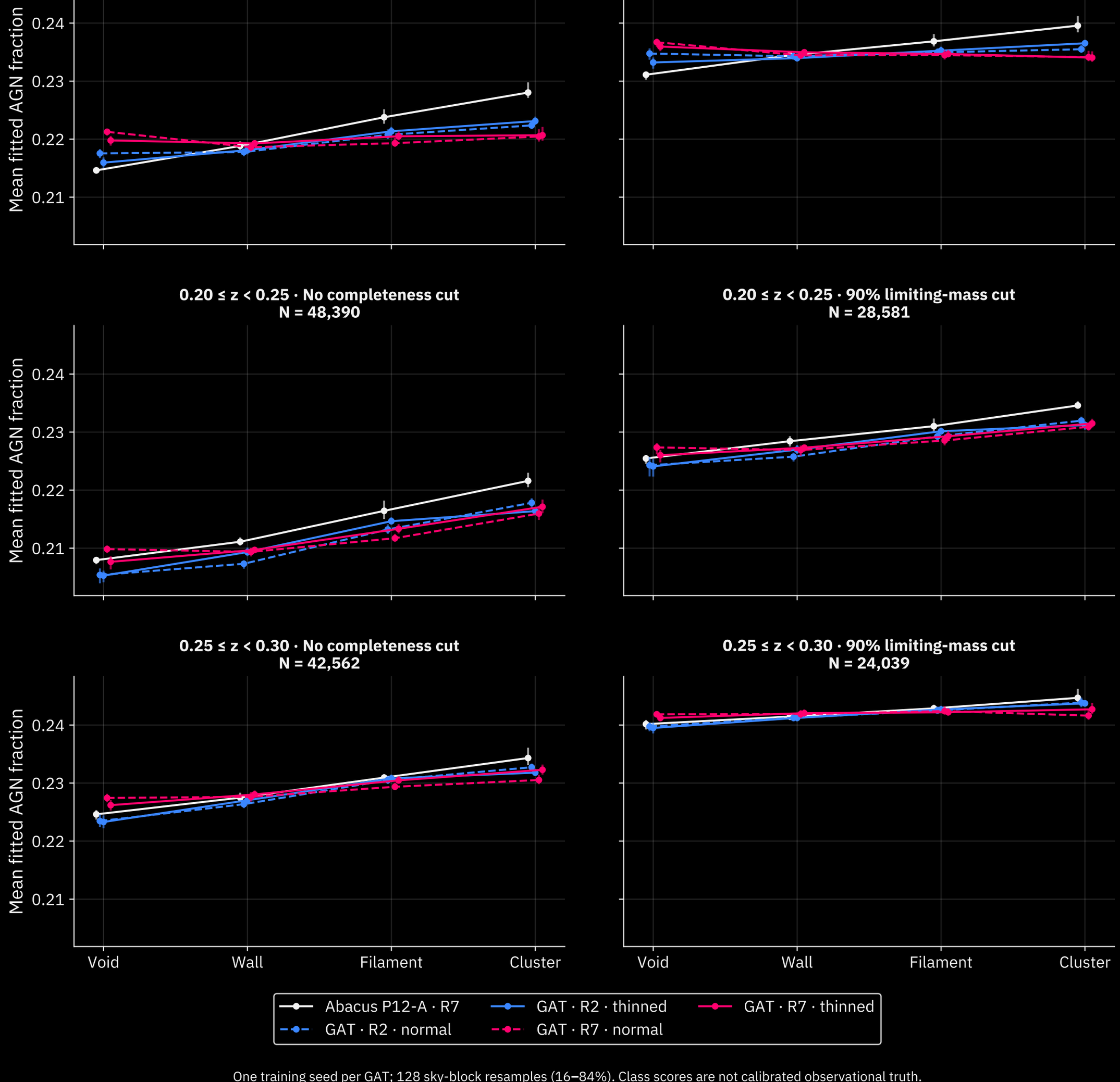
0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620

0.20 ≤ z < 0.25 · No completeness cut
N = 48,390

0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581

0.25 ≤ z < 0.30 · No completeness cut
N = 42,562

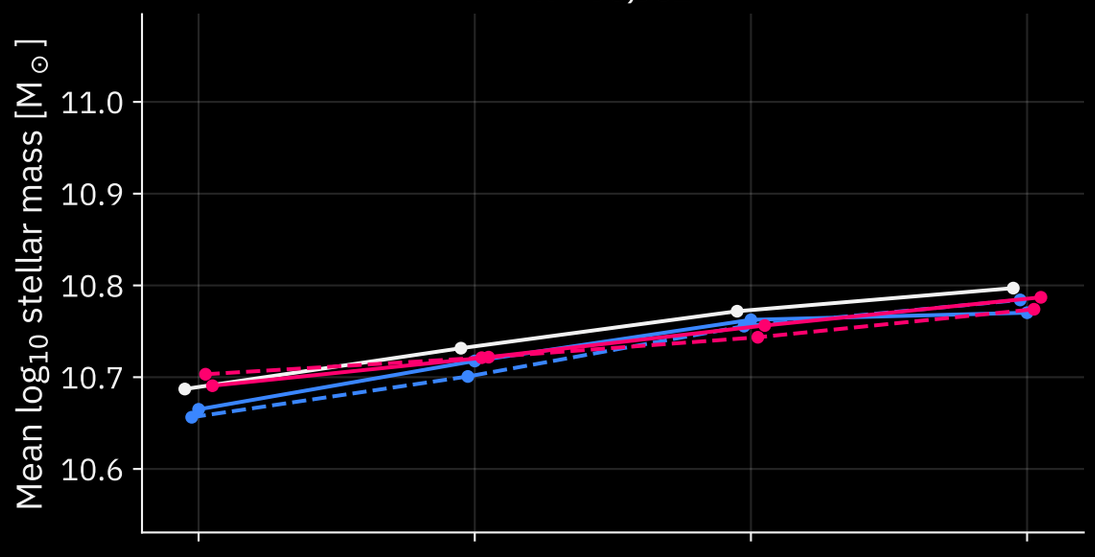
0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039



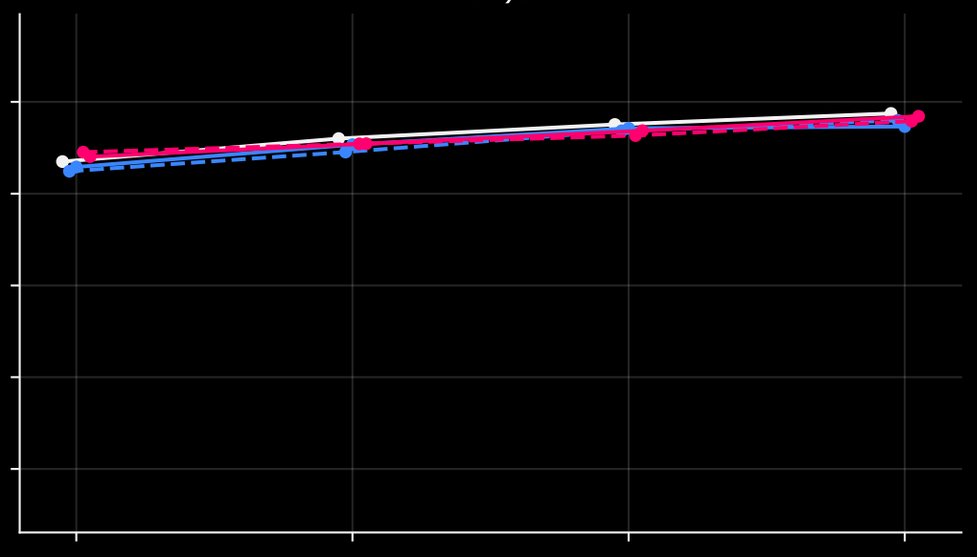
One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · mass
soft_controlled · Historical observed-feature normalization

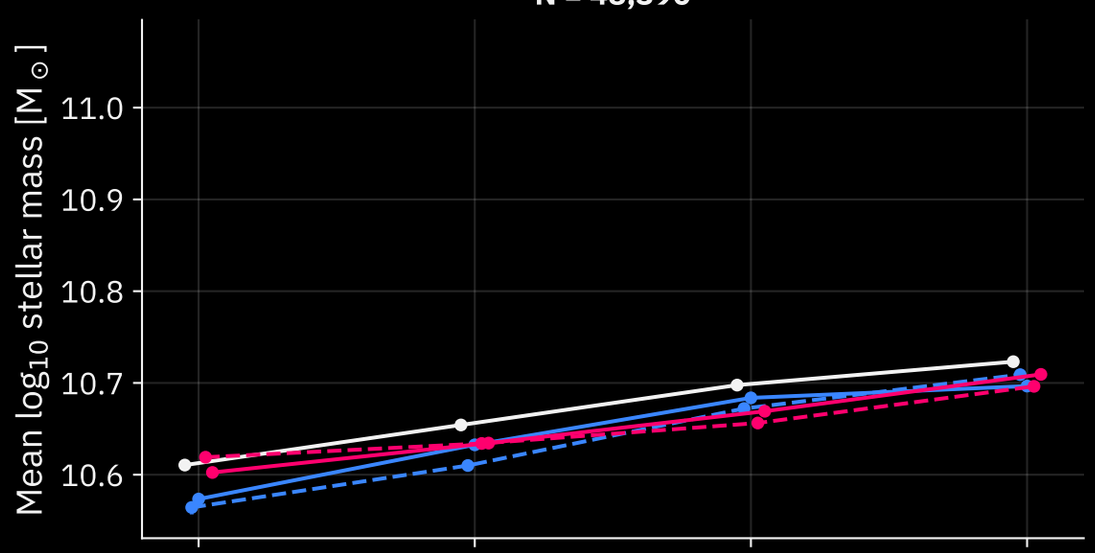
0.20 ≤ z < 0.30 · No completeness cut
N = 90,952



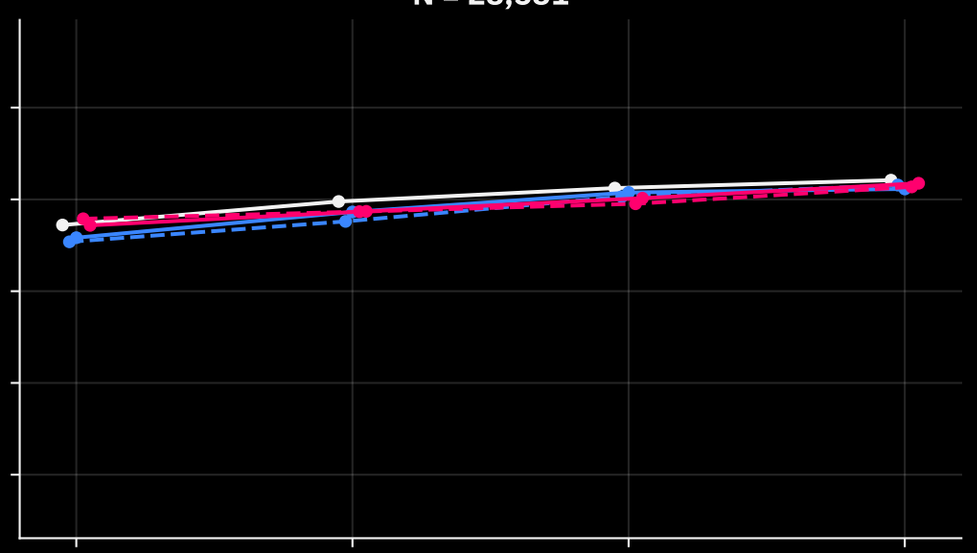
0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620



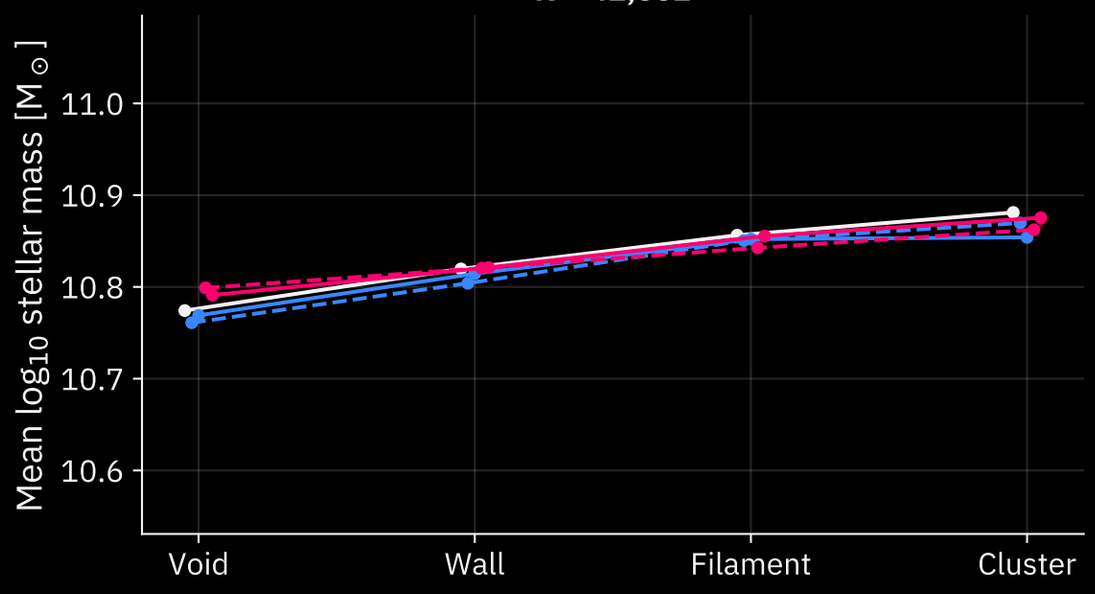
0.20 ≤ z < 0.25 · No completeness cut
N = 48,390



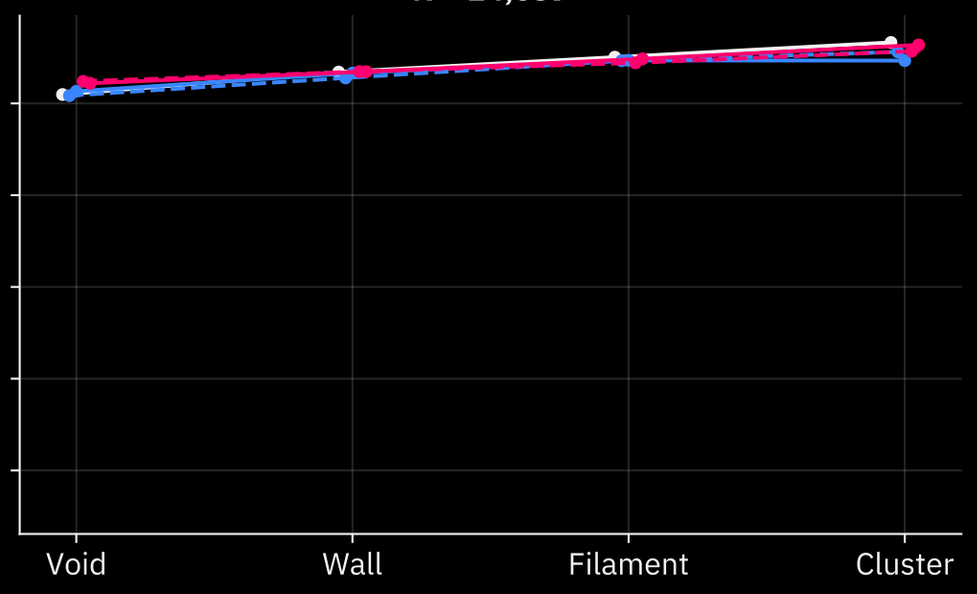
0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581



0.25 ≤ z < 0.30 · No completeness cut
N = 42,562



0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039

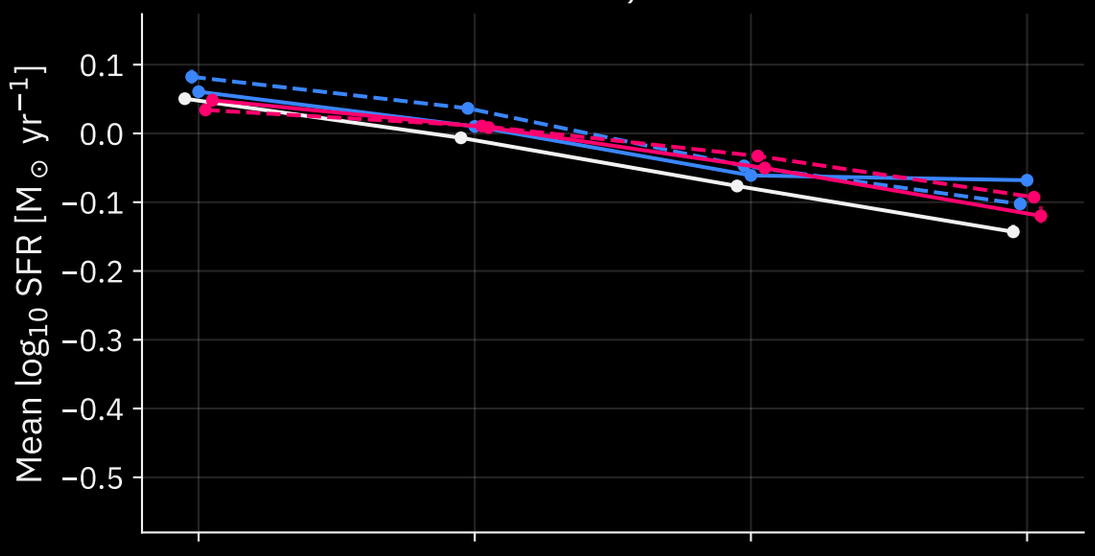


Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

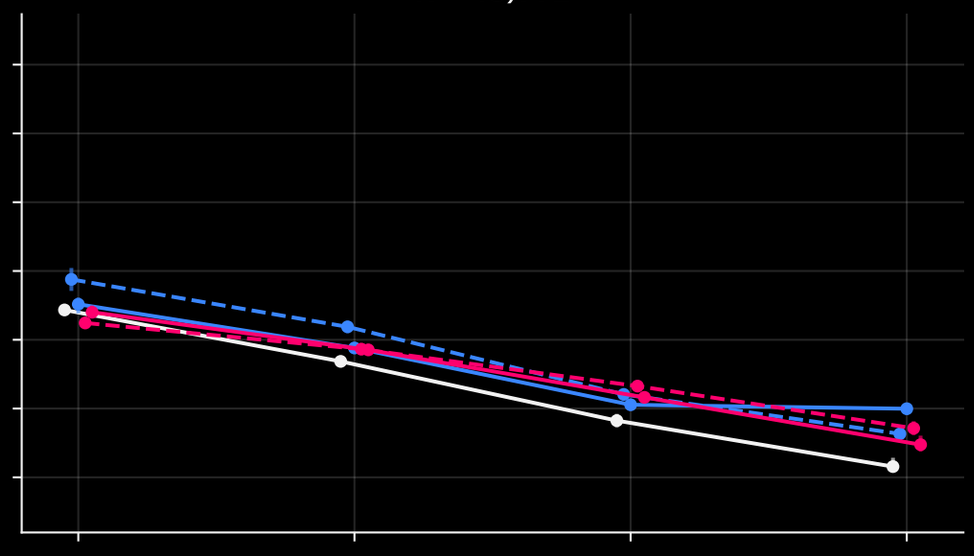
One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · sfr
soft_controlled · Historical observed-feature normalization

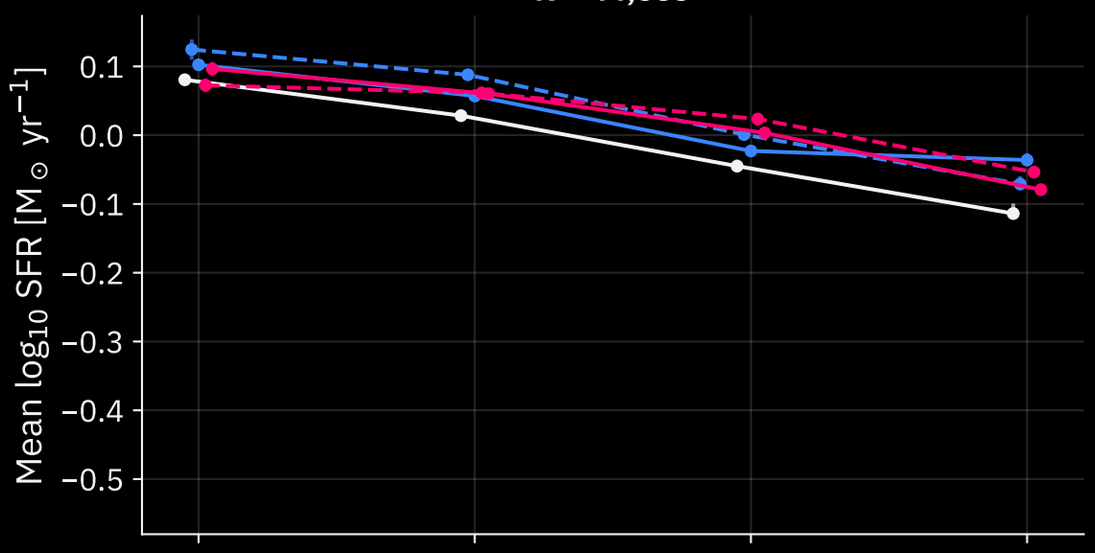
0.20 ≤ z < 0.30 · No completeness cut
N = 83,439



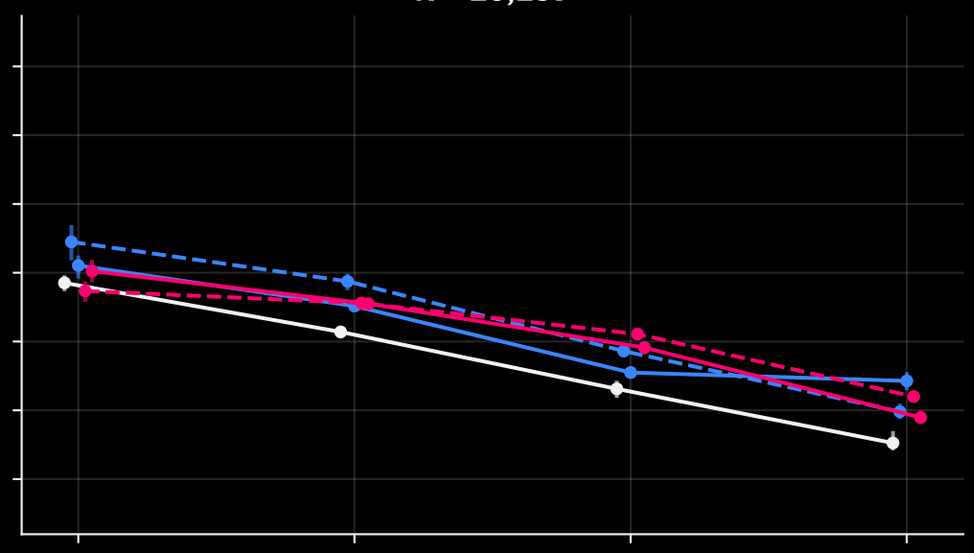
0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 48,772



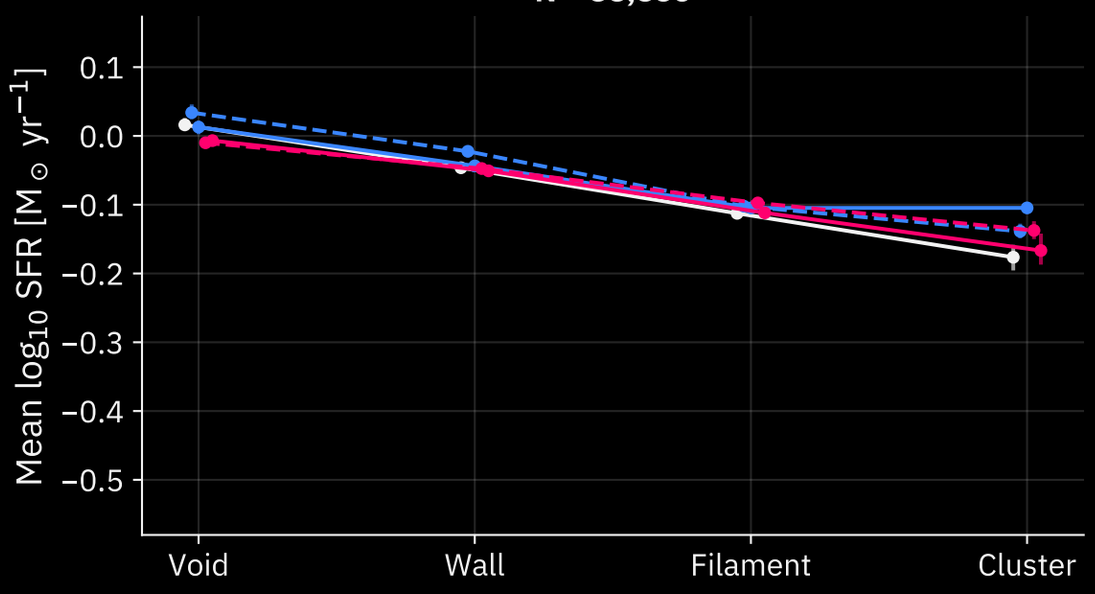
0.20 ≤ z < 0.25 · No completeness cut
N = 44,583



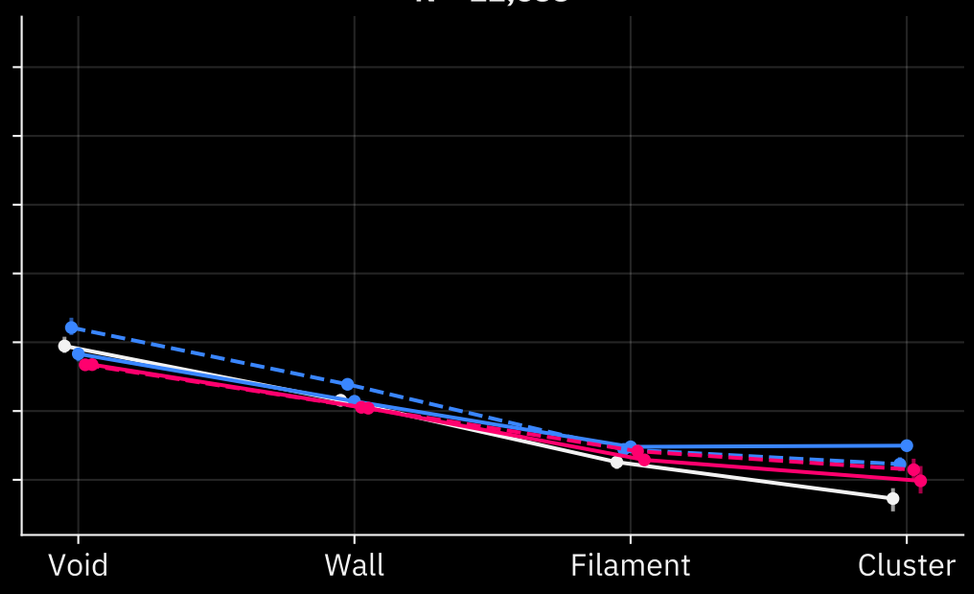
0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 26,239



0.25 ≤ z < 0.30 · No completeness cut
N = 38,856



0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 22,533

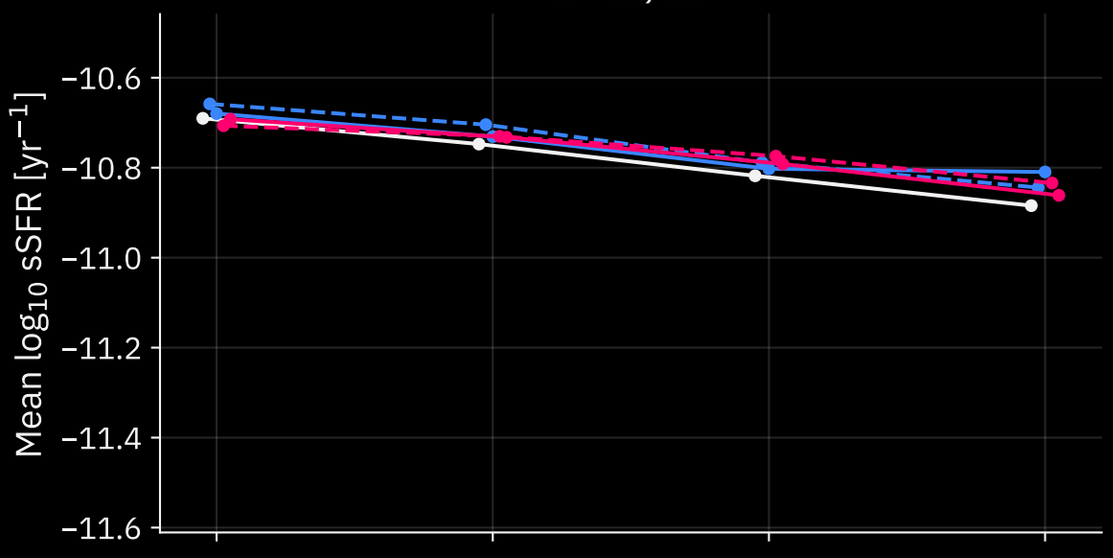


Abacus P12-A · R7
GAT · R2 · normal
GAT · R2 · thinned
GAT · R7 · normal
GAT · R7 · thinned

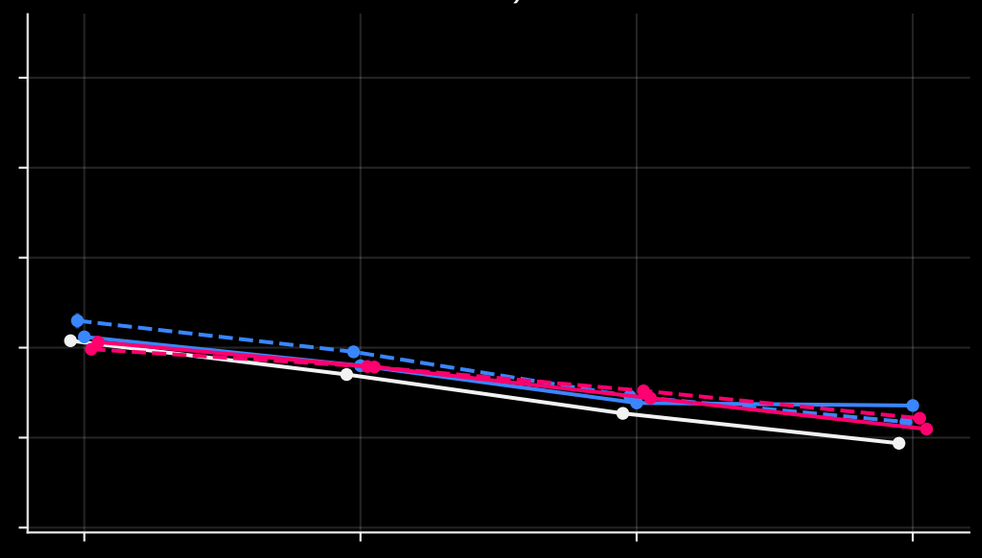
One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · ssfr
soft_controlled · Historical observed-feature normalization

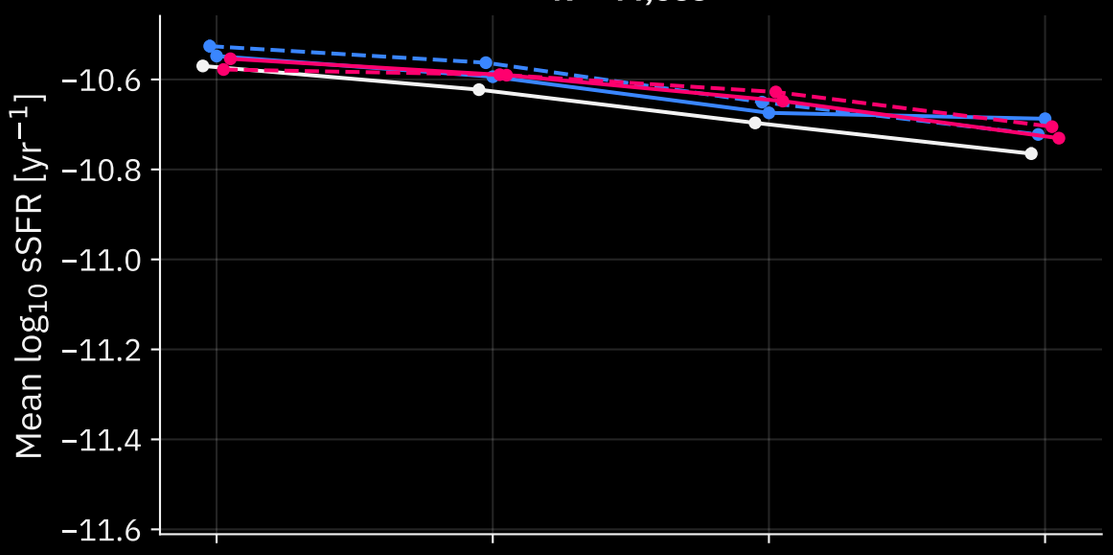
0.20 ≤ z < 0.30 · No completeness cut
N = 83,439



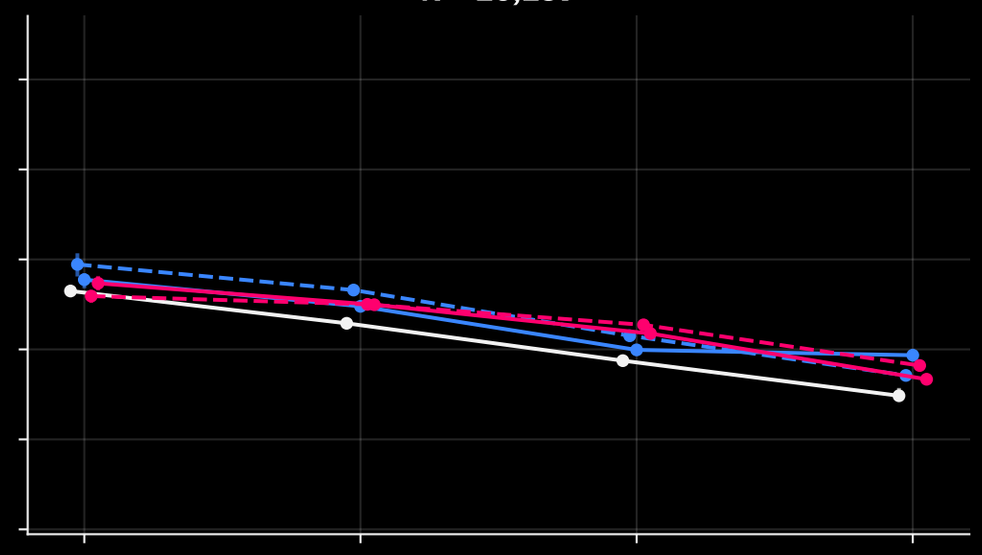
0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 48,772



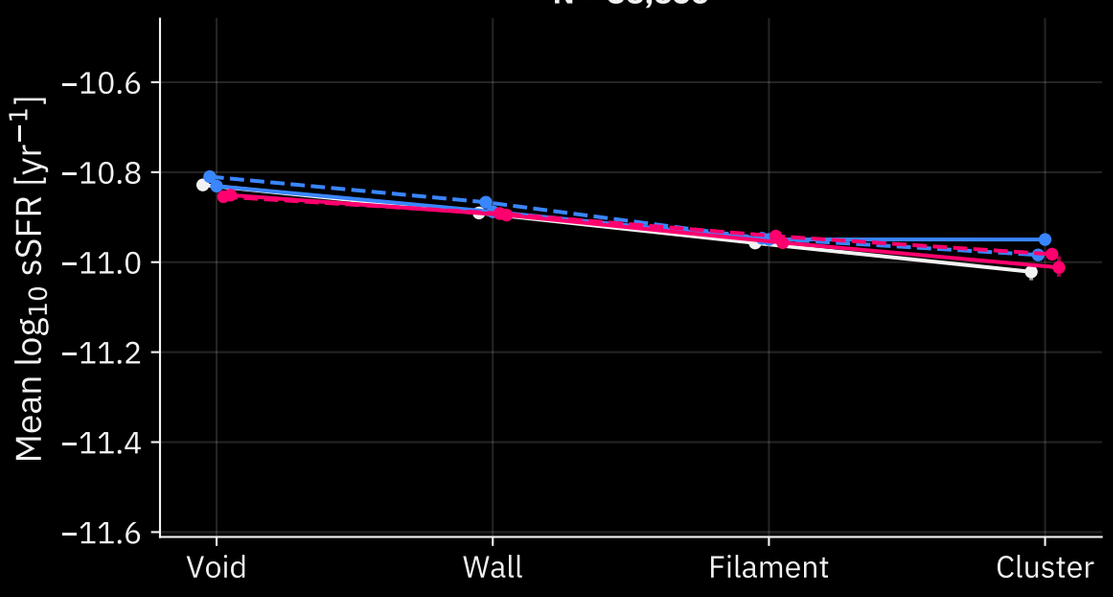
0.20 ≤ z < 0.25 · No completeness cut
N = 44,583



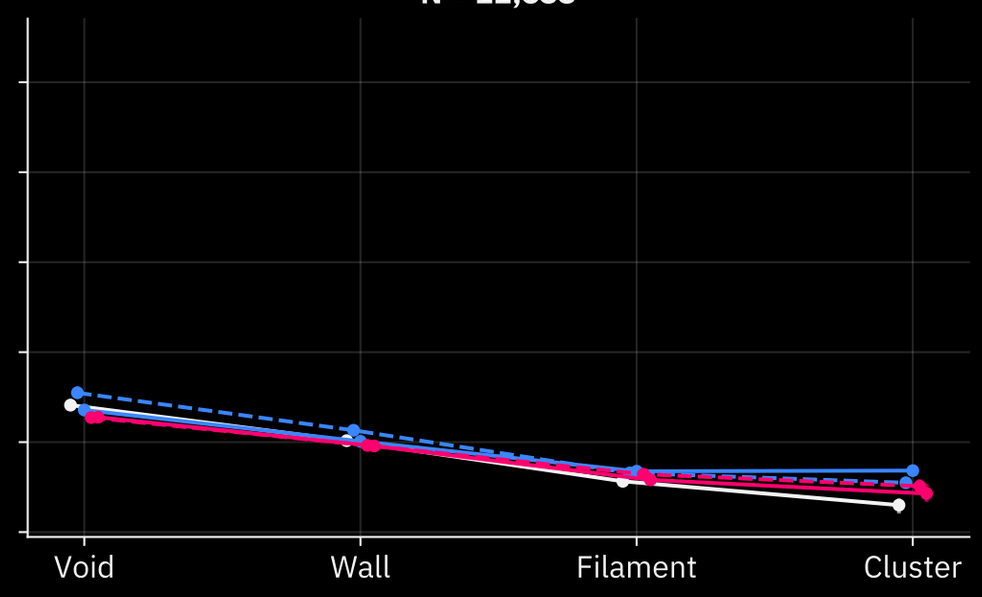
0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 26,239



0.25 ≤ z < 0.30 · No completeness cut
N = 38,856



0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 22,533

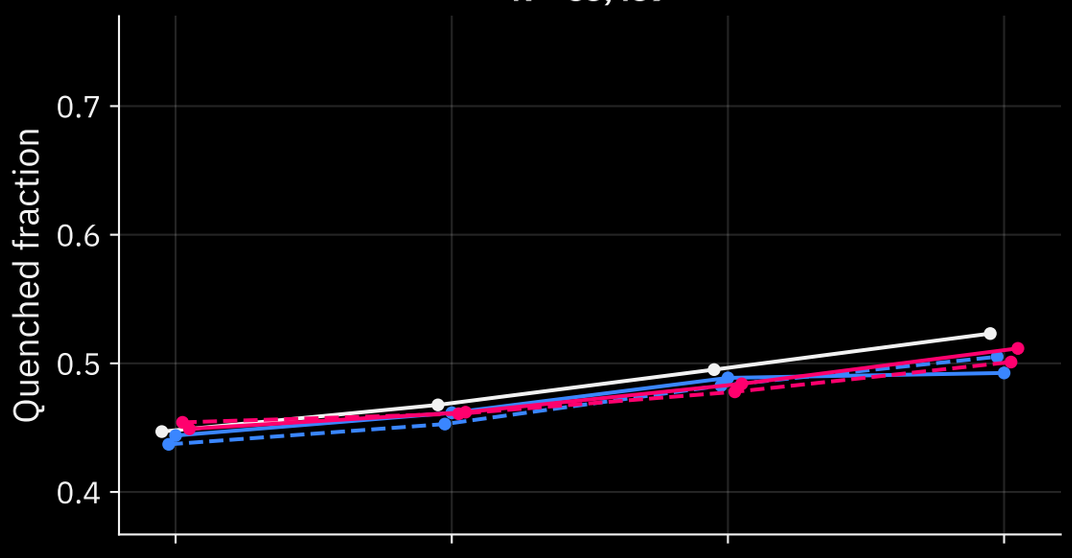


Abacus P12-A · R7
GAT · R2 · normal
GAT · R2 · thinned
GAT · R7 · normal
GAT · R7 · thinned

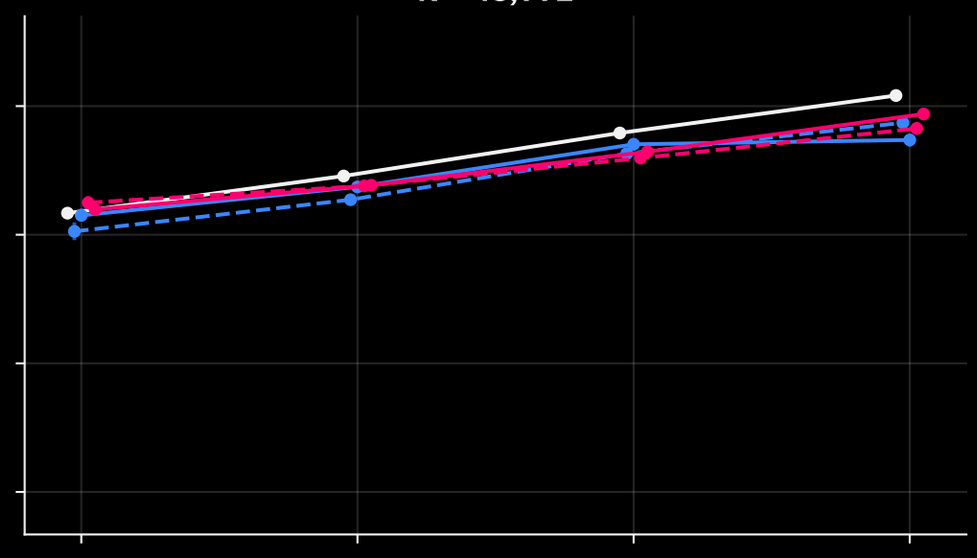
One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · quenched
soft_controlled · Historical observed-feature normalization

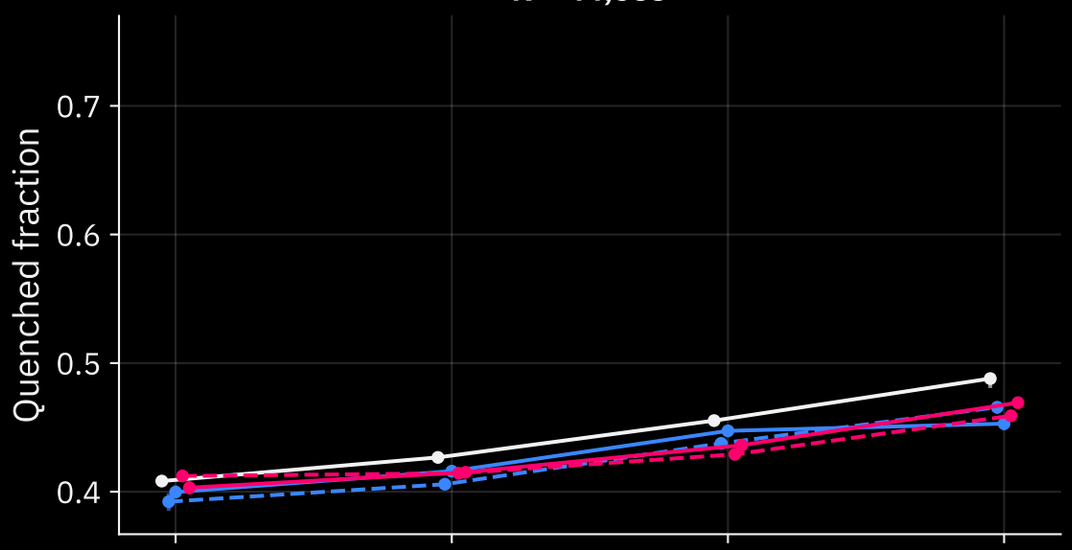
0.20 ≤ z < 0.30 · No completeness cut
N = 83,439



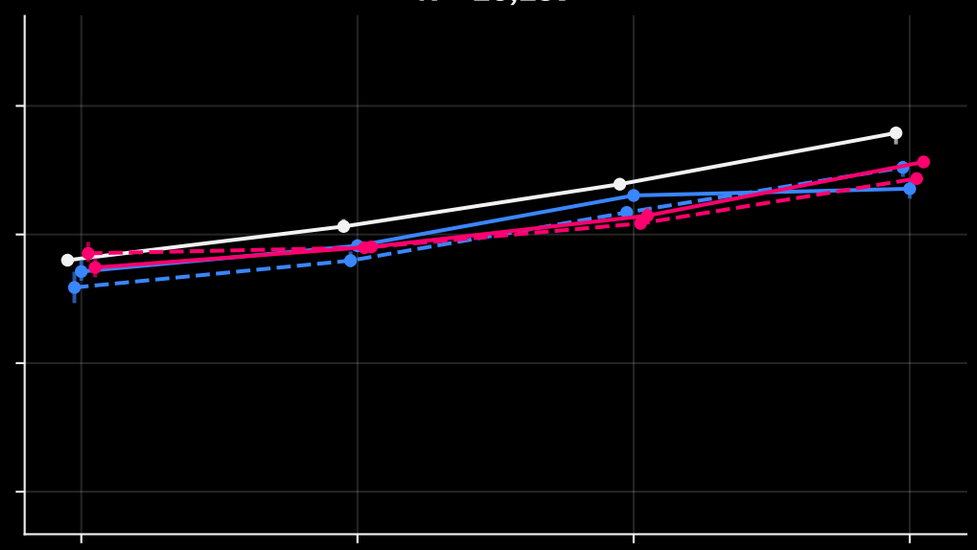
0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 48,772



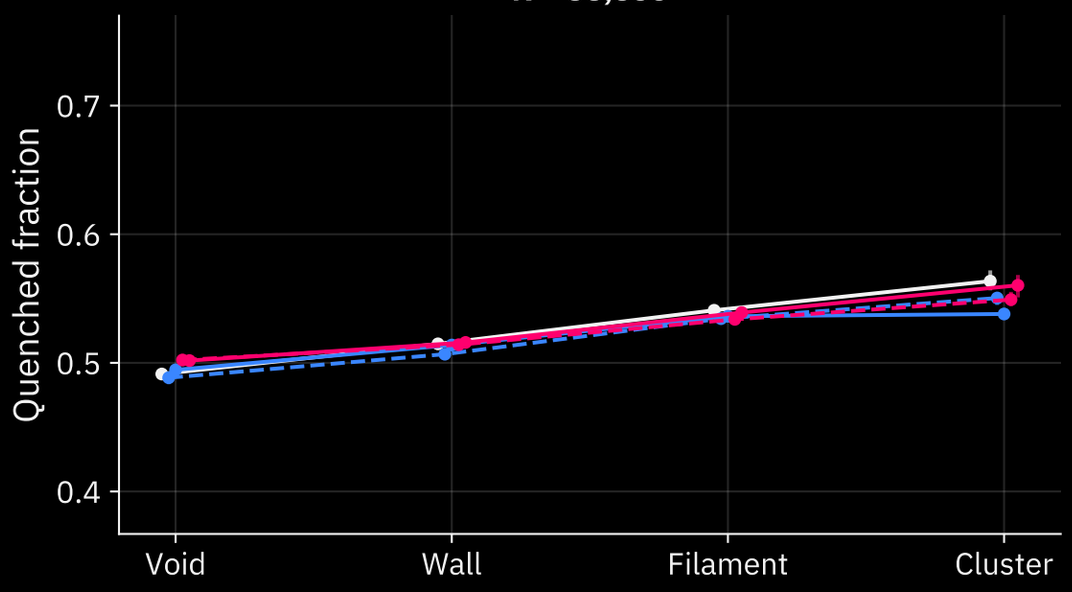
0.20 ≤ z < 0.25 · No completeness cut
N = 44,583



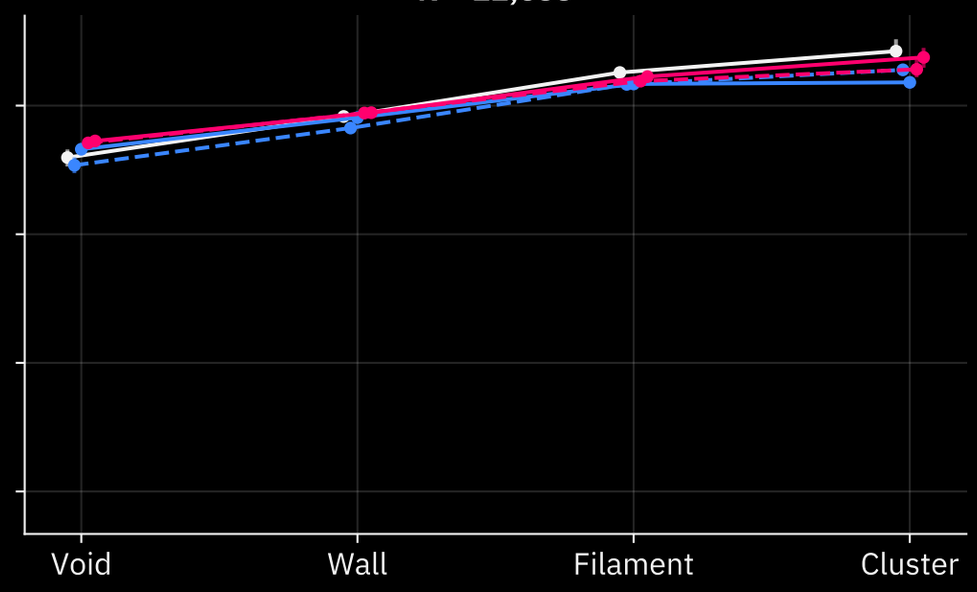
0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 26,239



0.25 ≤ z < 0.30 · No completeness cut
N = 38,856



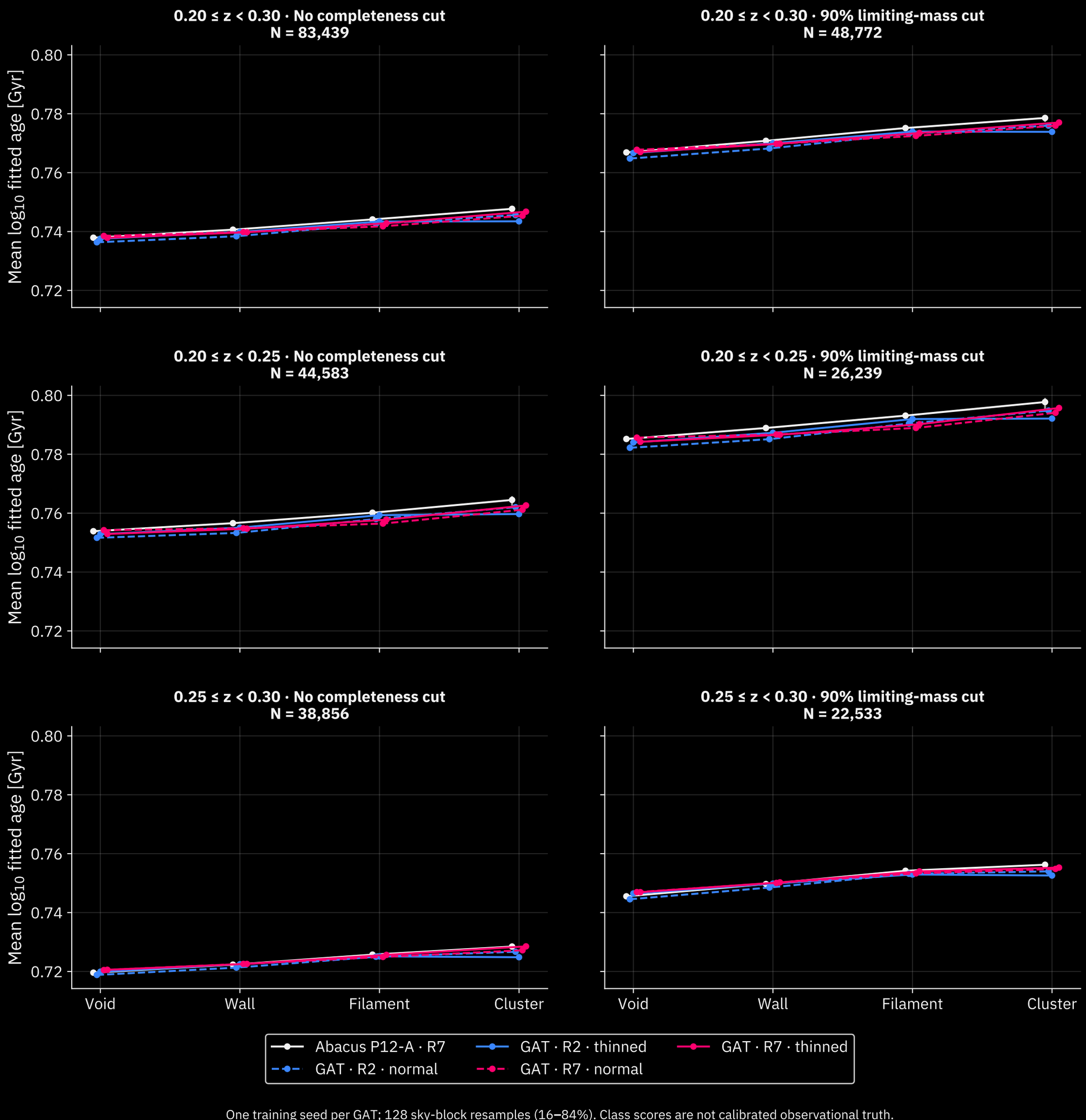
0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 22,533



Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · age
soft_controlled · Historical observed-feature normalization



One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · av
soft_controlled · Historical observed-feature normalization

0.20 ≤ z < 0.30 · No completeness cut
N = 83,439

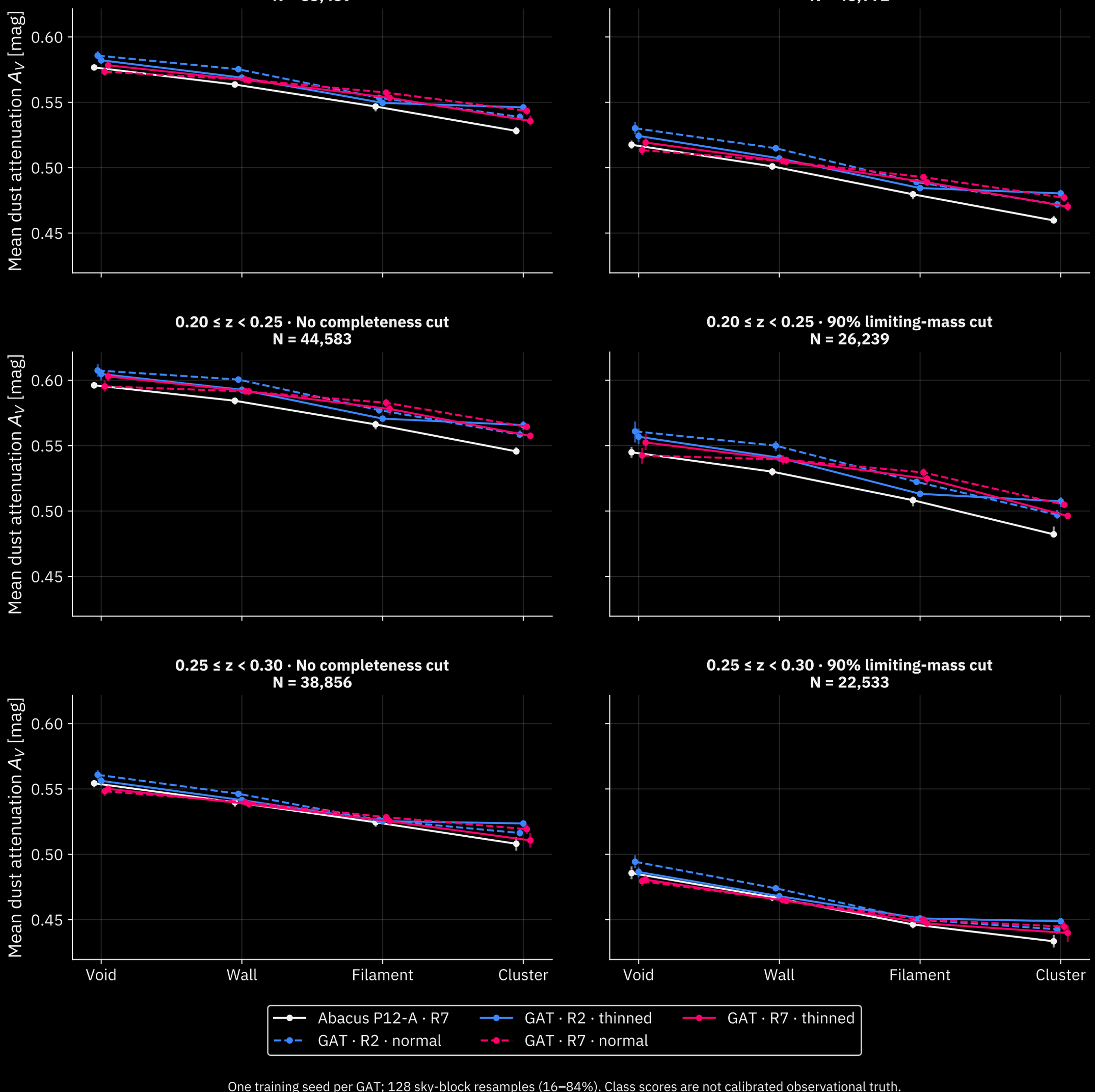
0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 48,772

0.20 ≤ z < 0.25 · No completeness cut
N = 44,583

0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 26,239

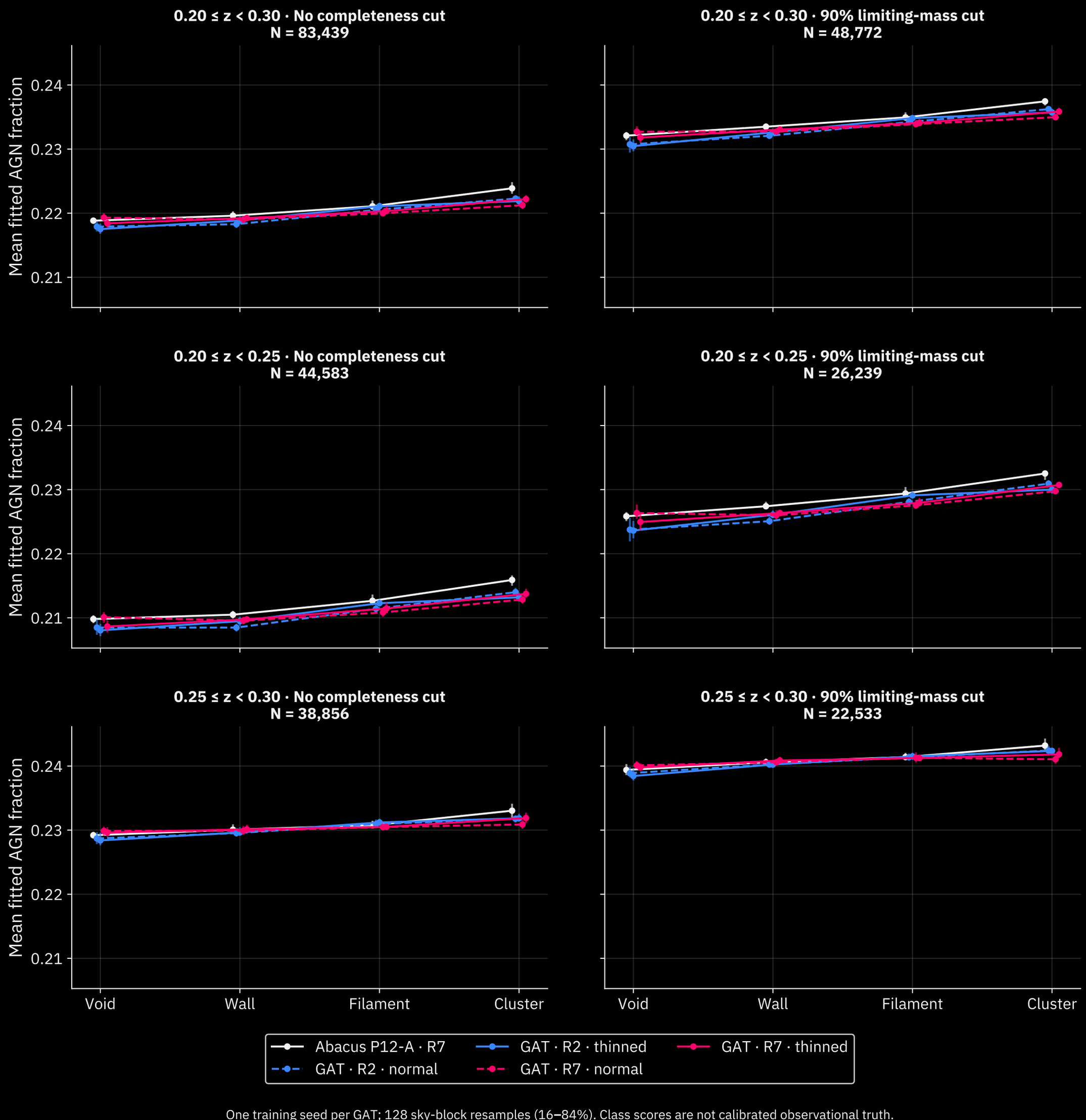
0.25 ≤ z < 0.30 · No completeness cut
N = 38,856

0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 22,533



One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

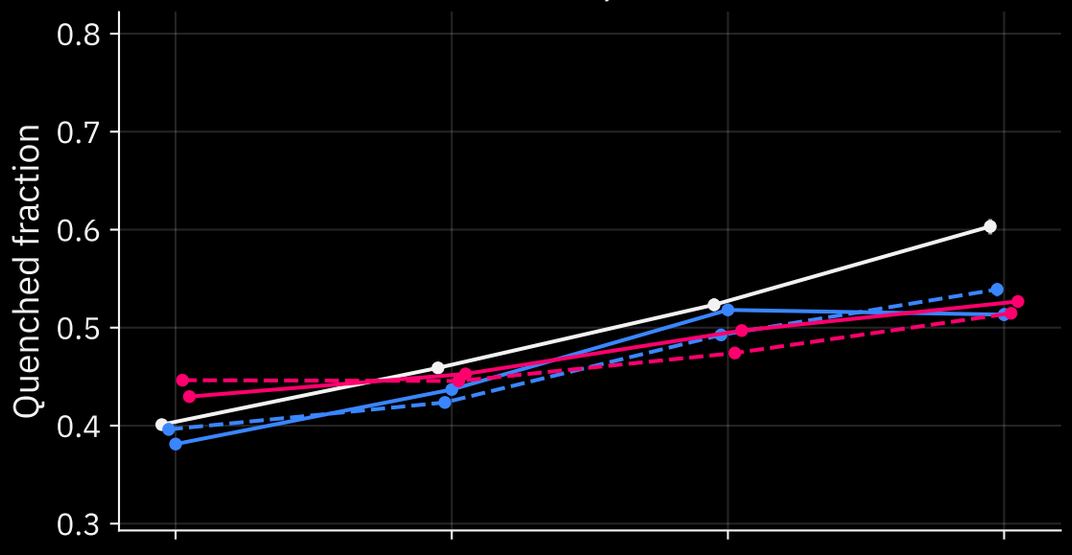
Actual TNG GAT versus Abacus P12-A · fagn
soft_controlled · Historical observed-feature normalization



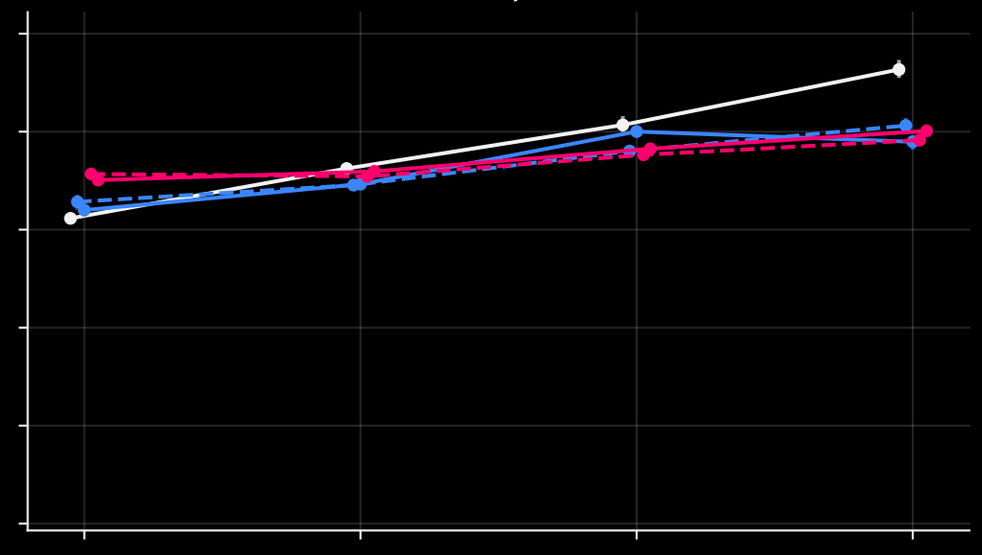
One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · quenched
hard_raw · Historical observed-feature normalization

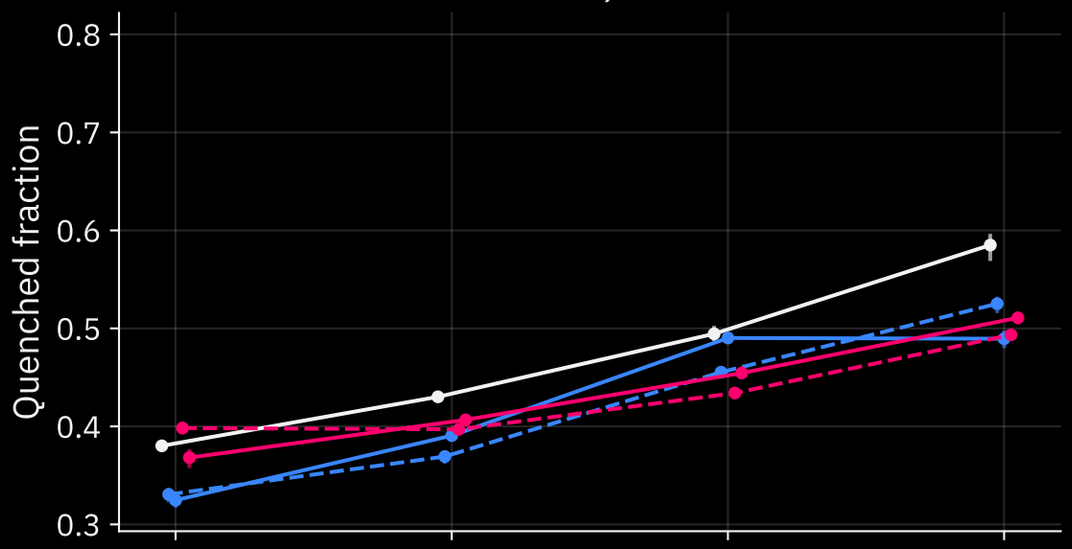
0.20 ≤ z < 0.30 · No completeness cut
N = 90,952



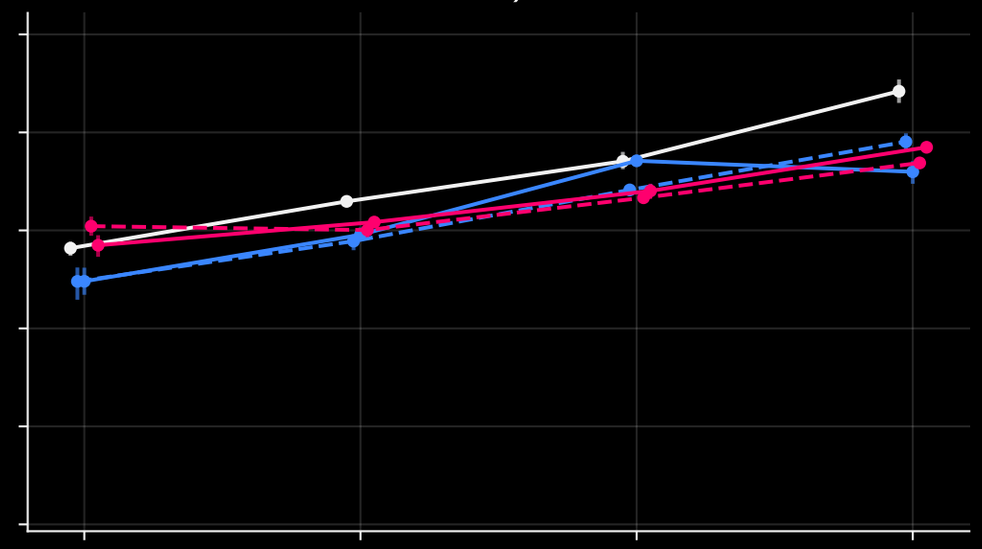
0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620



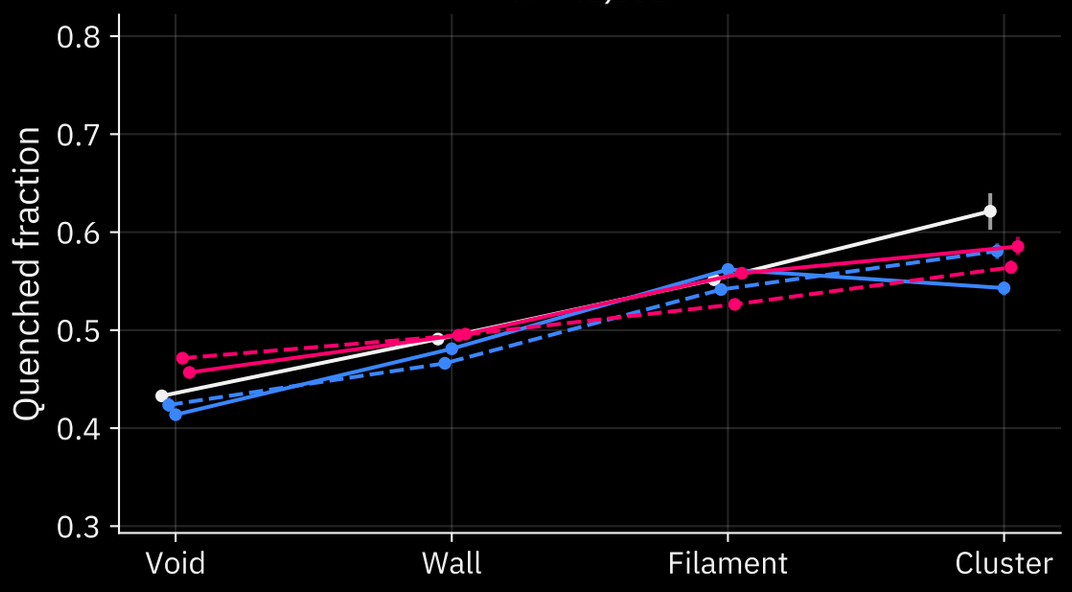
0.20 ≤ z < 0.25 · No completeness cut
N = 48,390



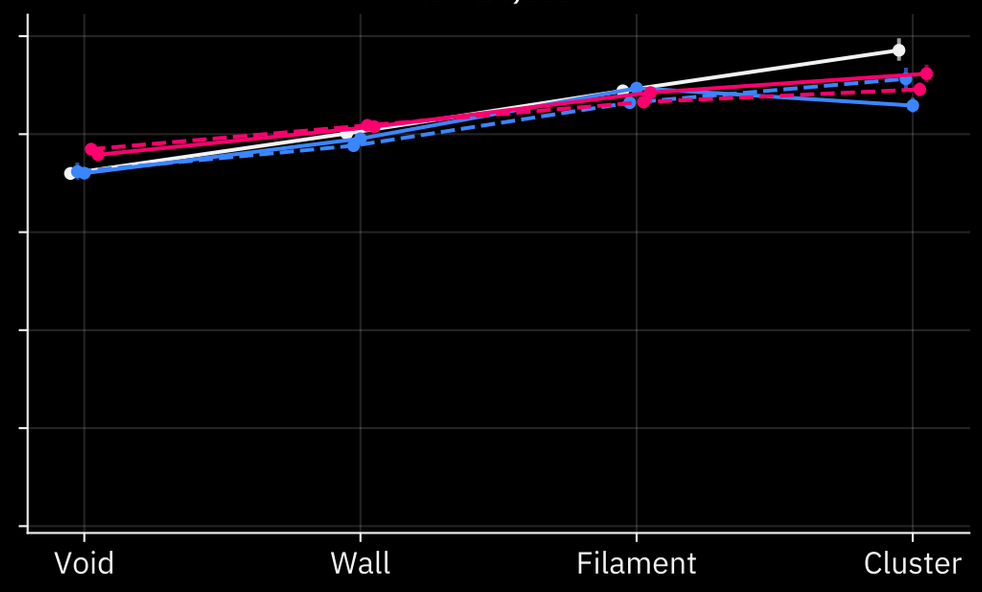
0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581



0.25 ≤ z < 0.30 · No completeness cut
N = 42,562



0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039

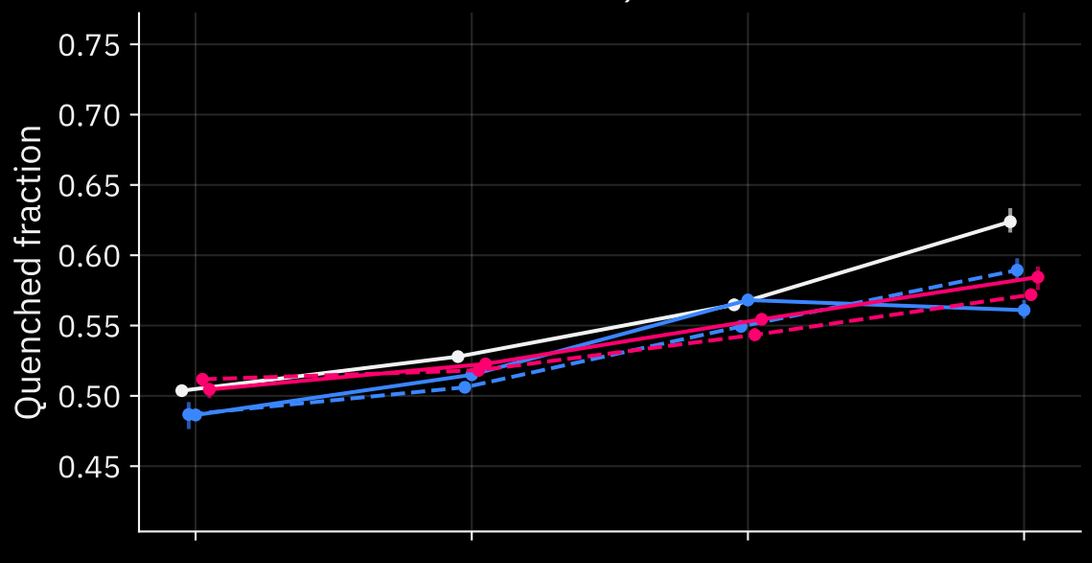


Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

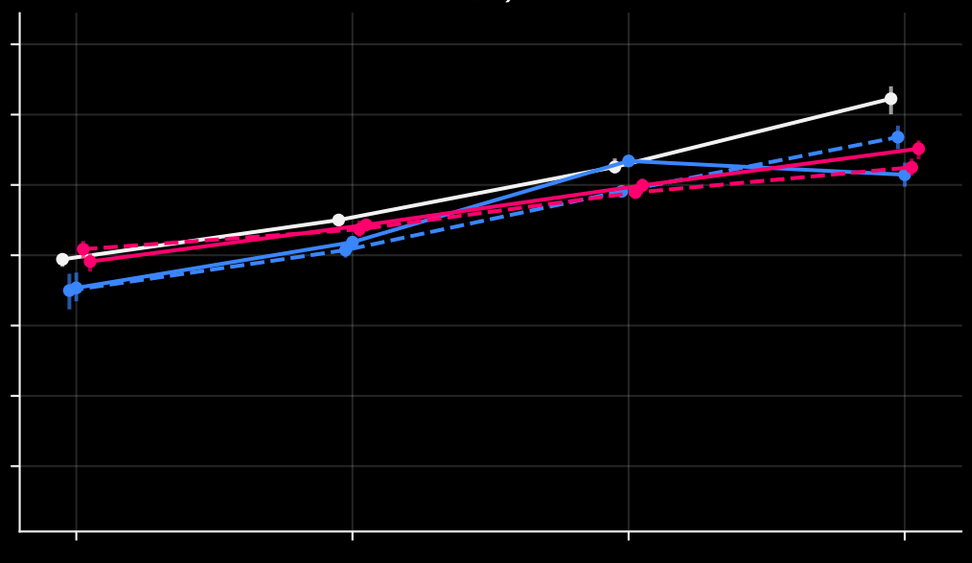
One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · quenched
hard_controlled · Historical observed-feature normalization

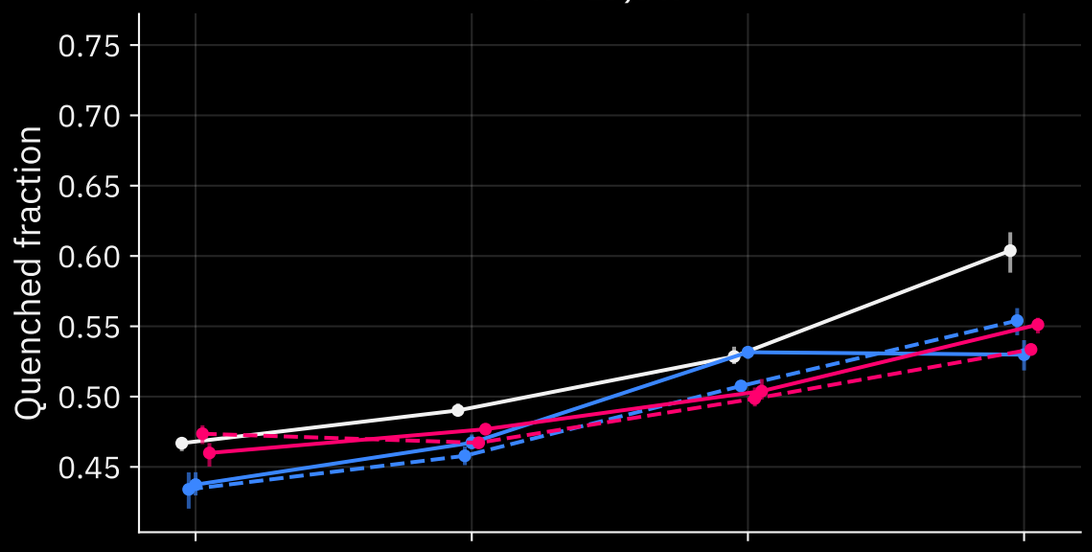
0.20 ≤ z < 0.30 · No completeness cut
N = 60,226



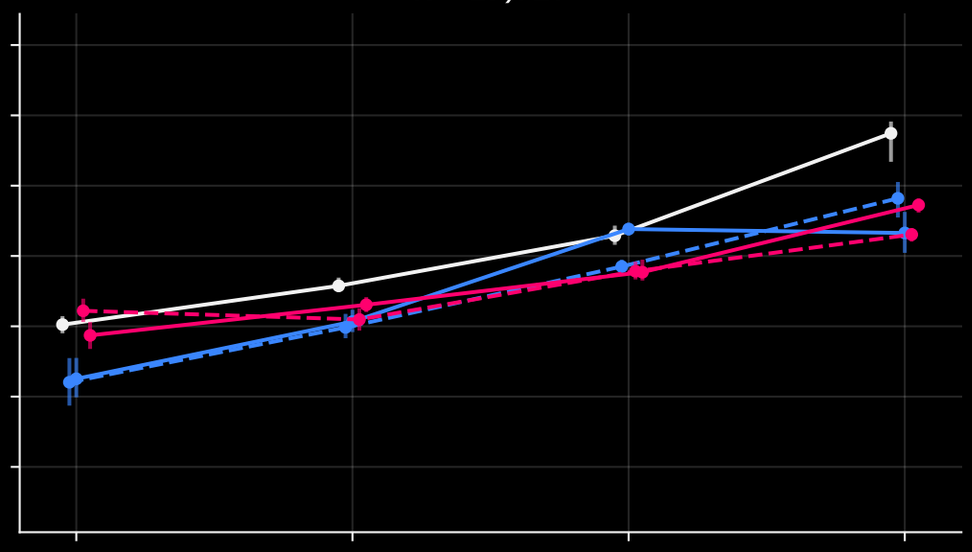
0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 39,446



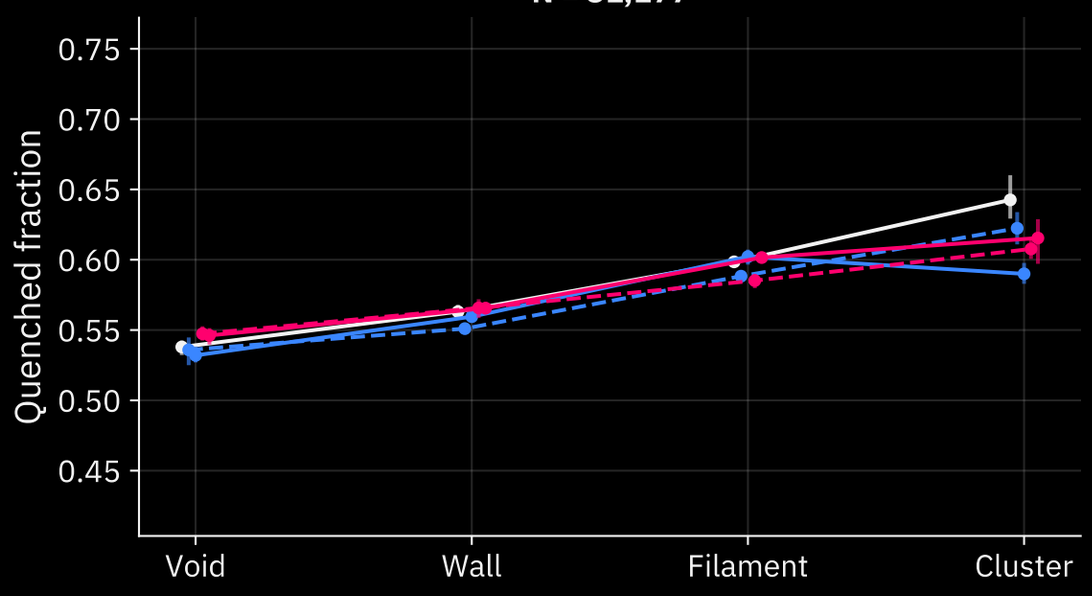
0.20 ≤ z < 0.25 · No completeness cut
N = 29,049



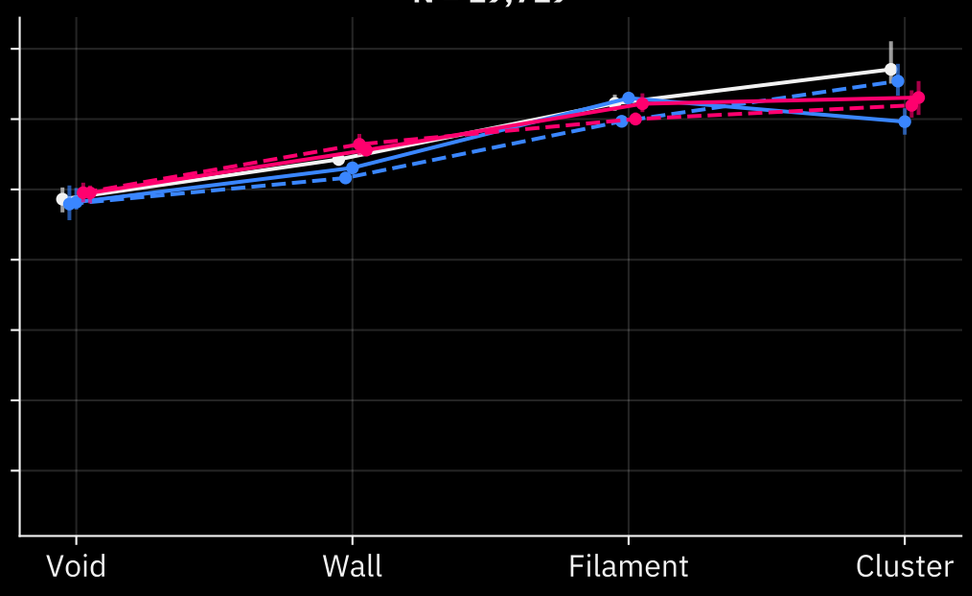
0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 19,727



0.25 ≤ z < 0.30 · No completeness cut
N = 31,177



0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 19,719

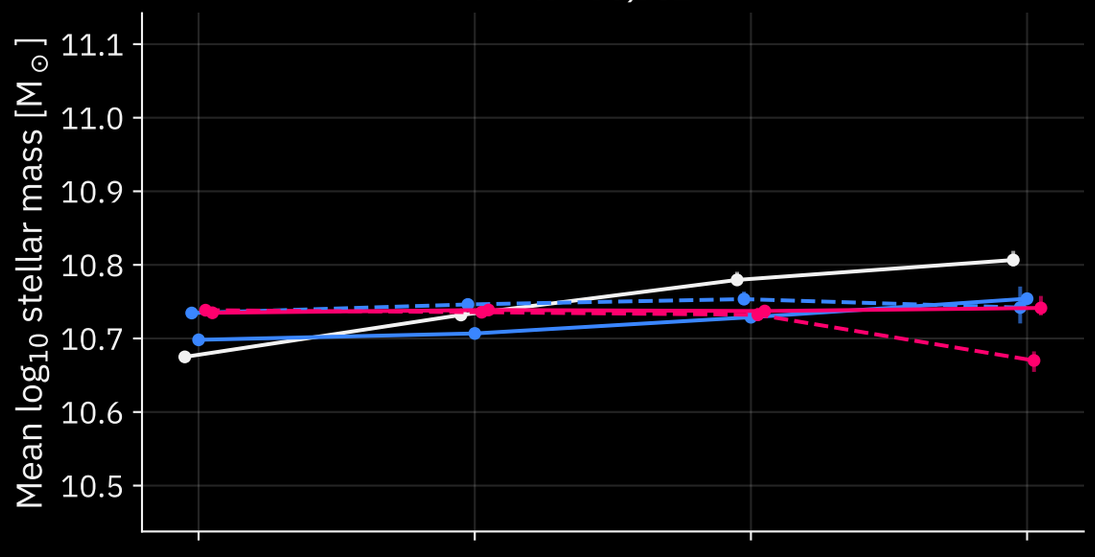


Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

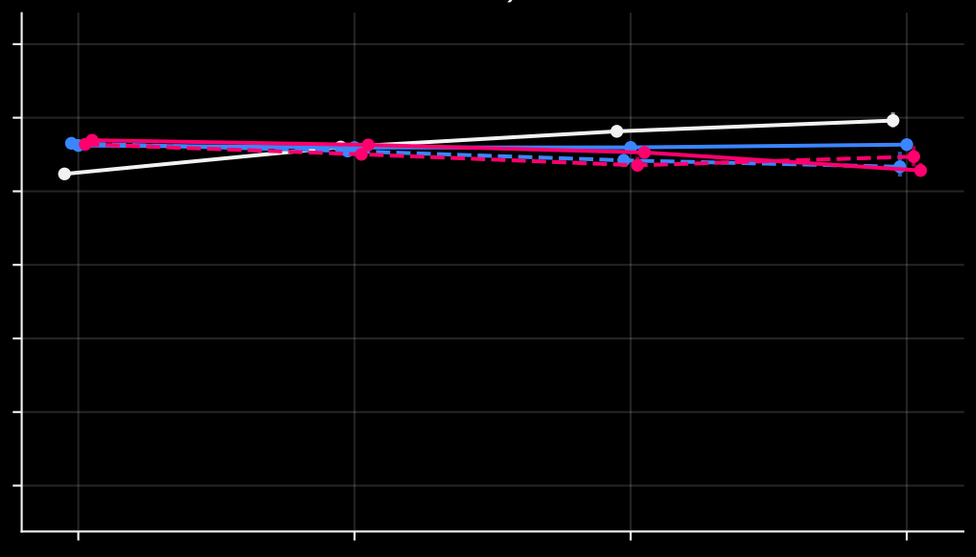
One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · mass
soft_raw · Frozen training-feature normalization

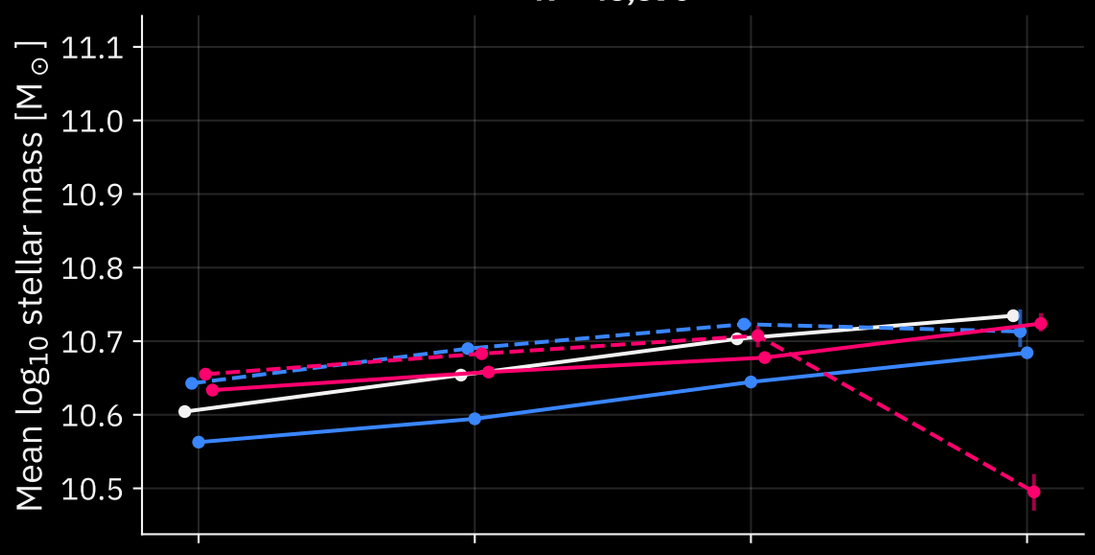
0.20 ≤ z < 0.30 · No completeness cut
N = 90,952



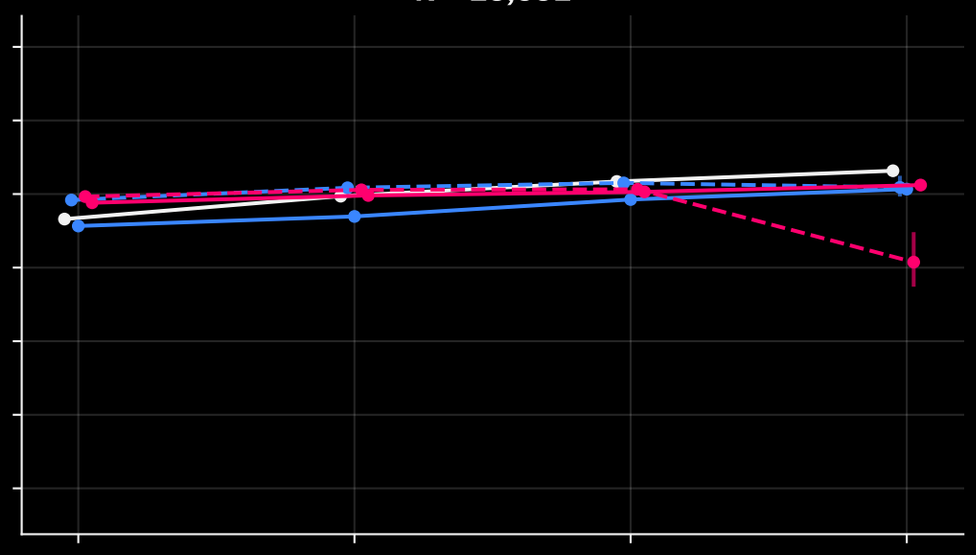
0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620



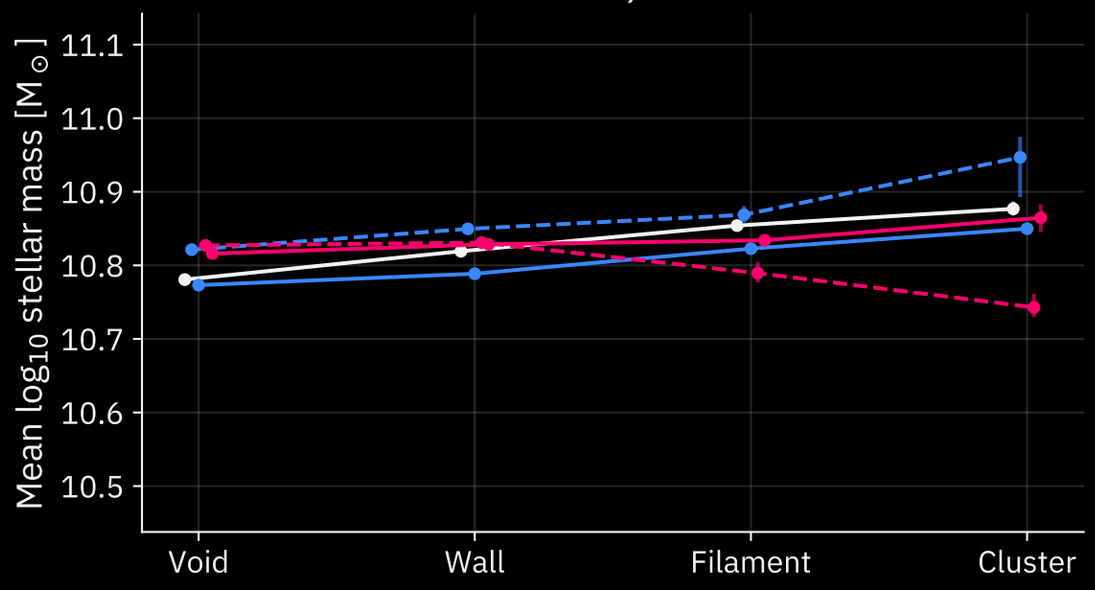
0.20 ≤ z < 0.25 · No completeness cut
N = 48,390



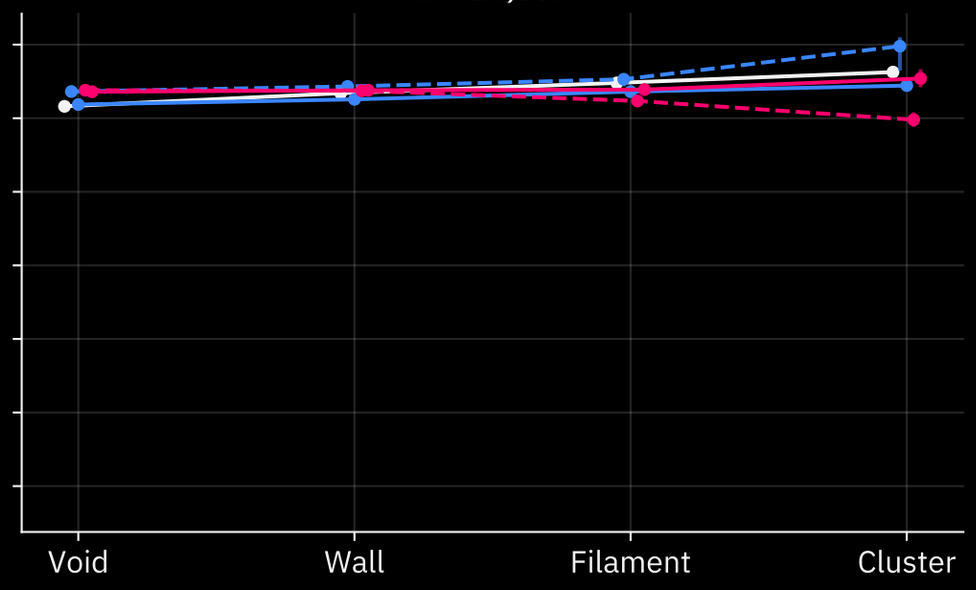
0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581



0.25 ≤ z < 0.30 · No completeness cut
N = 42,562



0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039

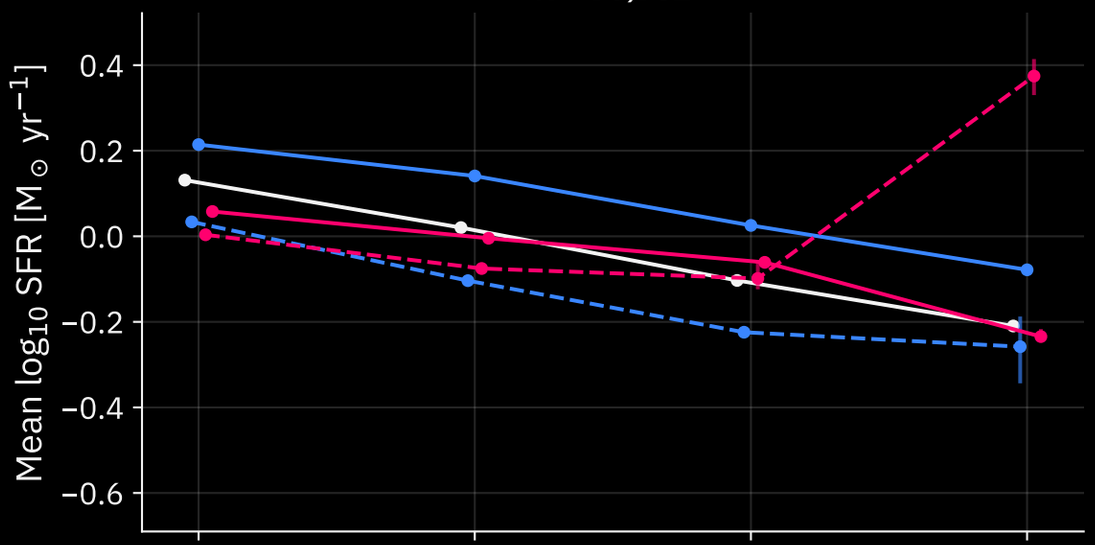


Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

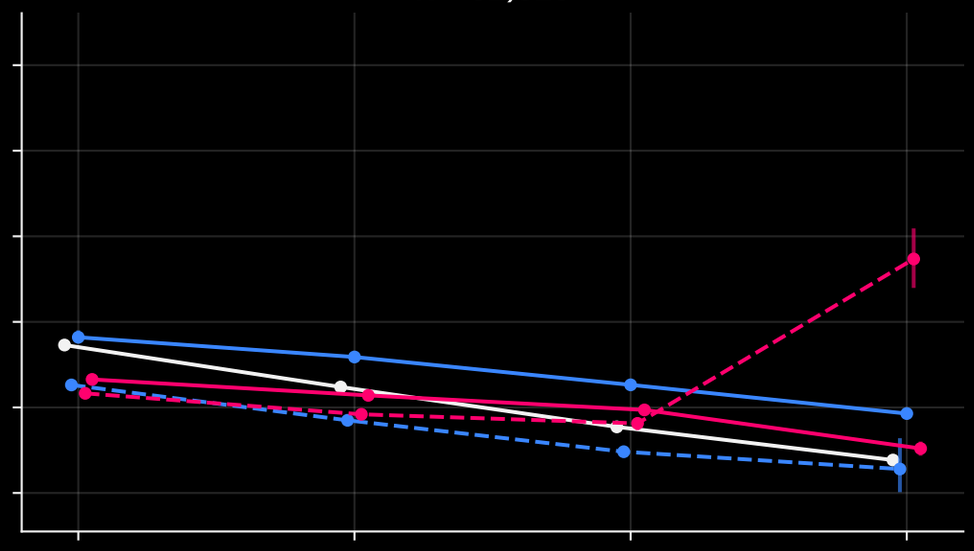
One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · sfr
soft_raw · Frozen training-feature normalization

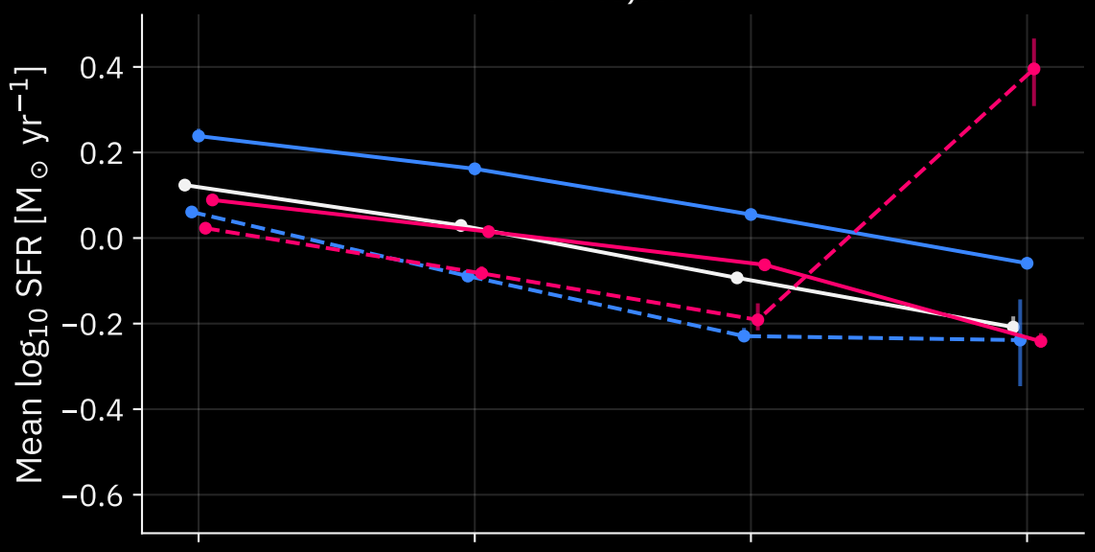
0.20 ≤ z < 0.30 · No completeness cut
N = 90,952



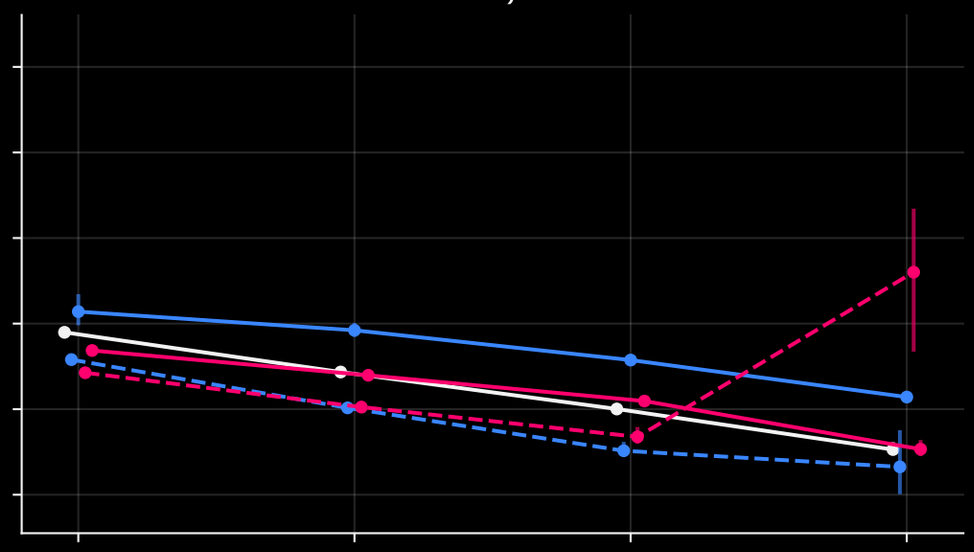
0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620



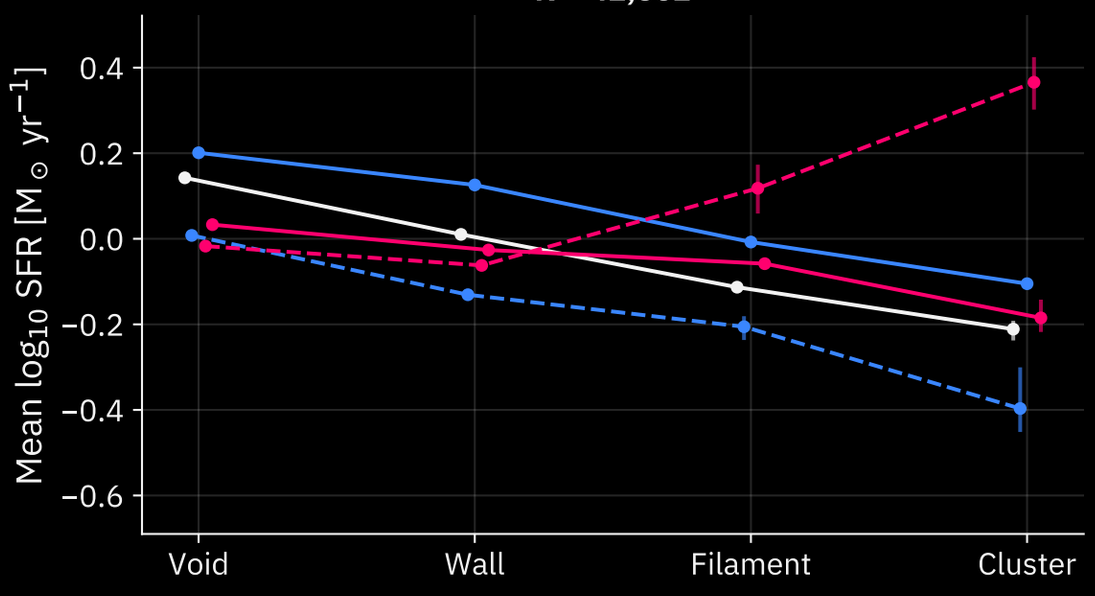
0.20 ≤ z < 0.25 · No completeness cut
N = 48,390



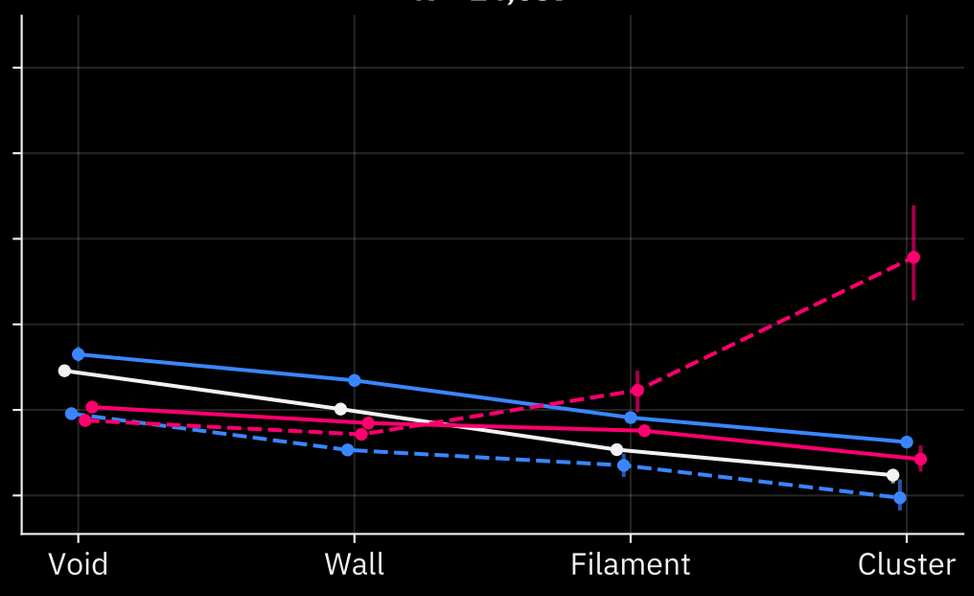
0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581



0.25 ≤ z < 0.30 · No completeness cut
N = 42,562



0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039

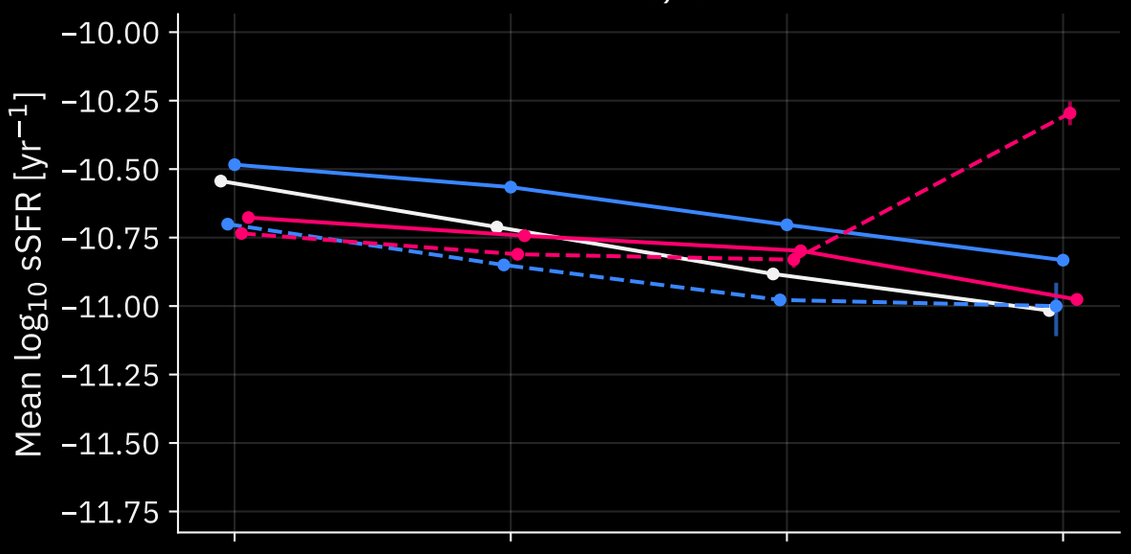


Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

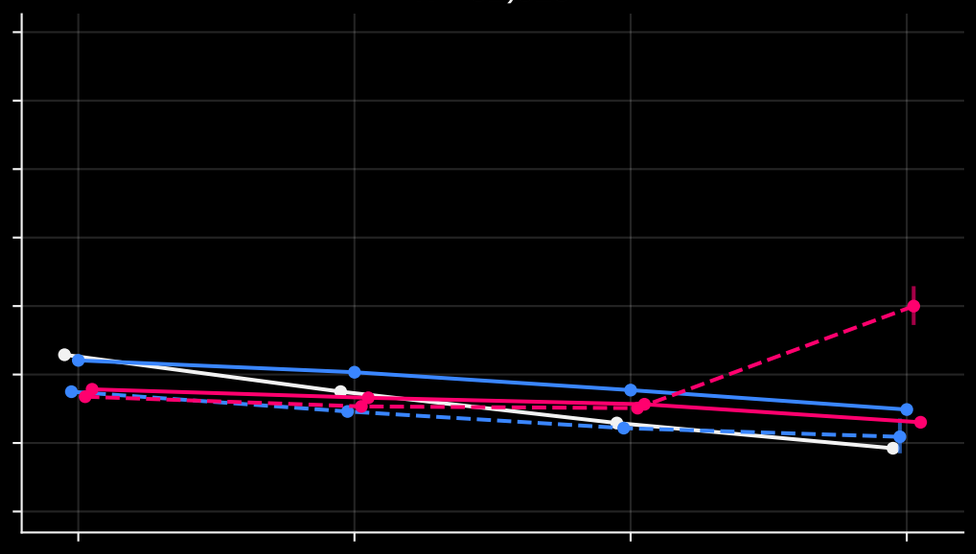
One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · ssfr
soft_raw · Frozen training-feature normalization

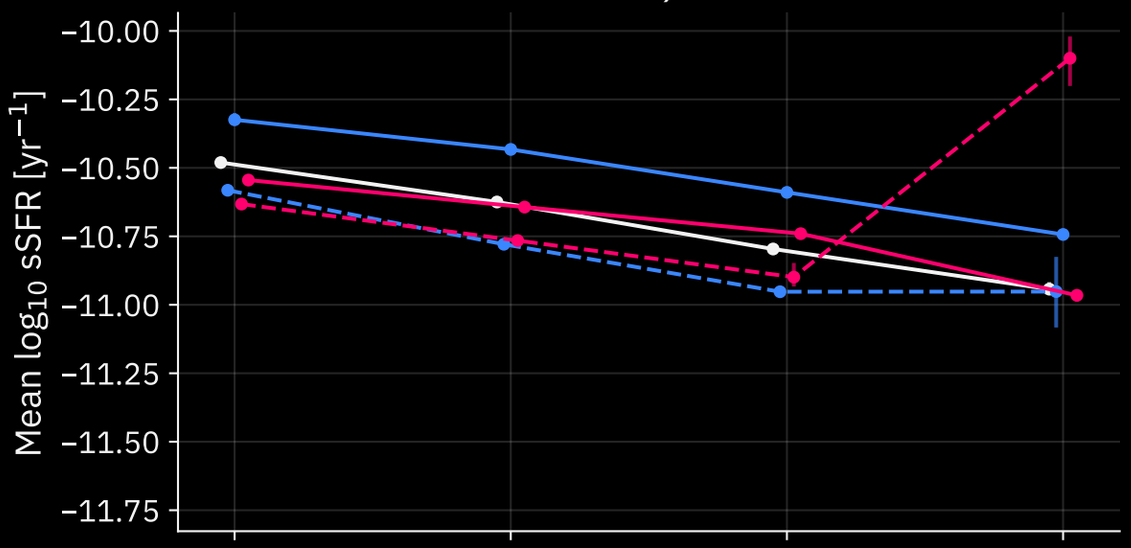
0.20 ≤ z < 0.30 · No completeness cut
N = 90,952



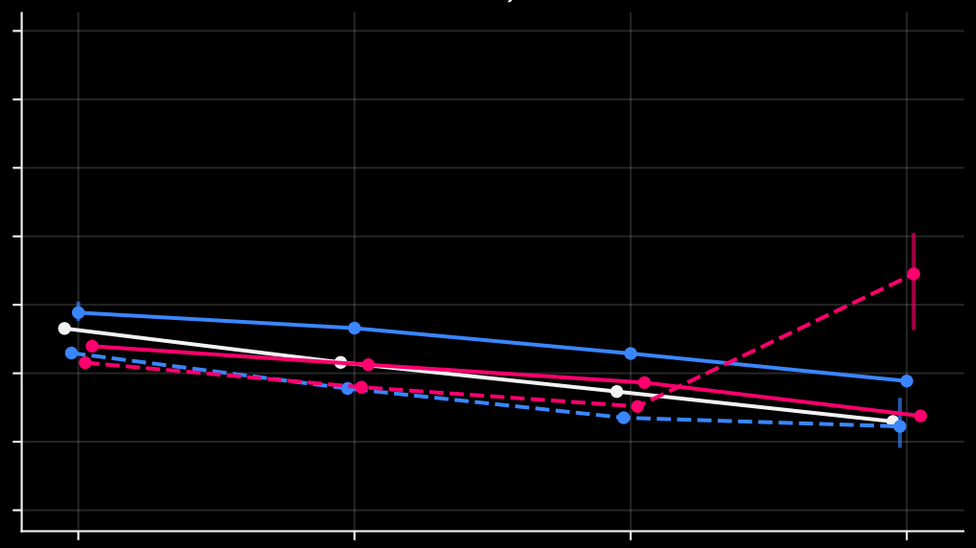
0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620



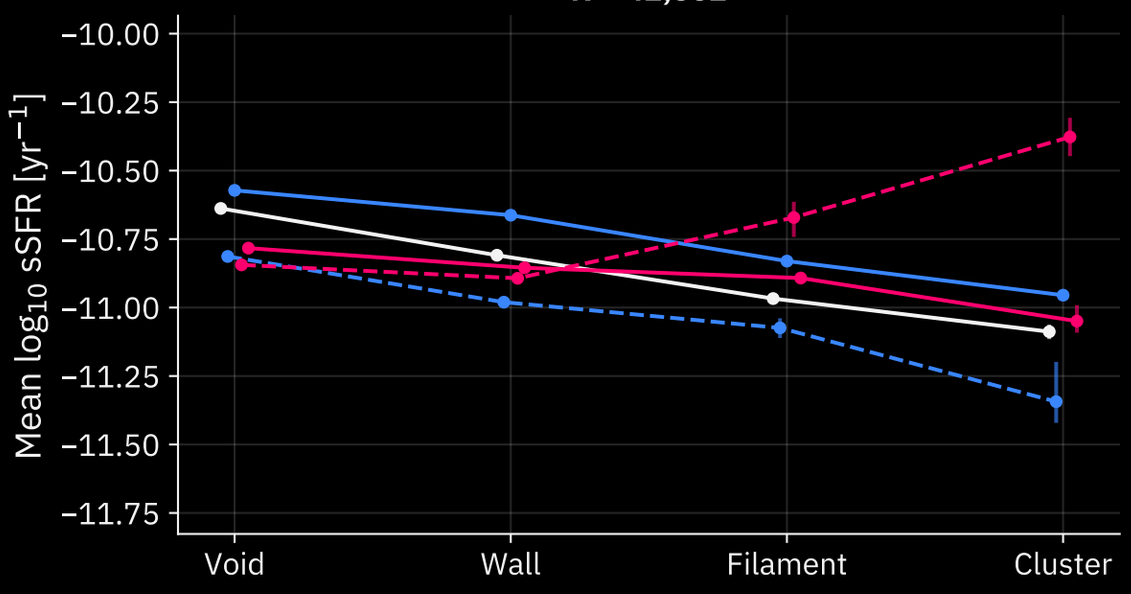
0.20 ≤ z < 0.25 · No completeness cut
N = 48,390



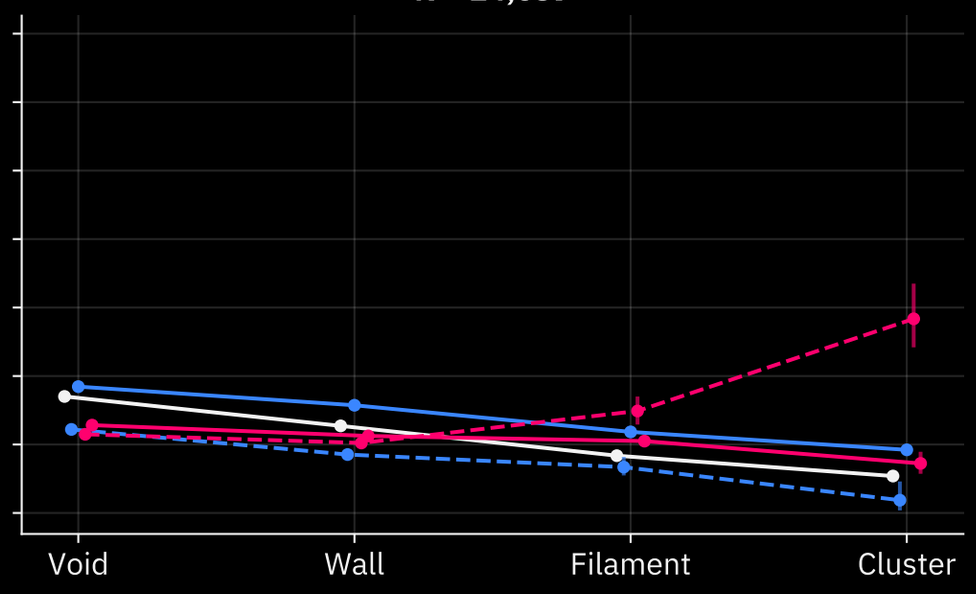
0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581



0.25 ≤ z < 0.30 · No completeness cut
N = 42,562



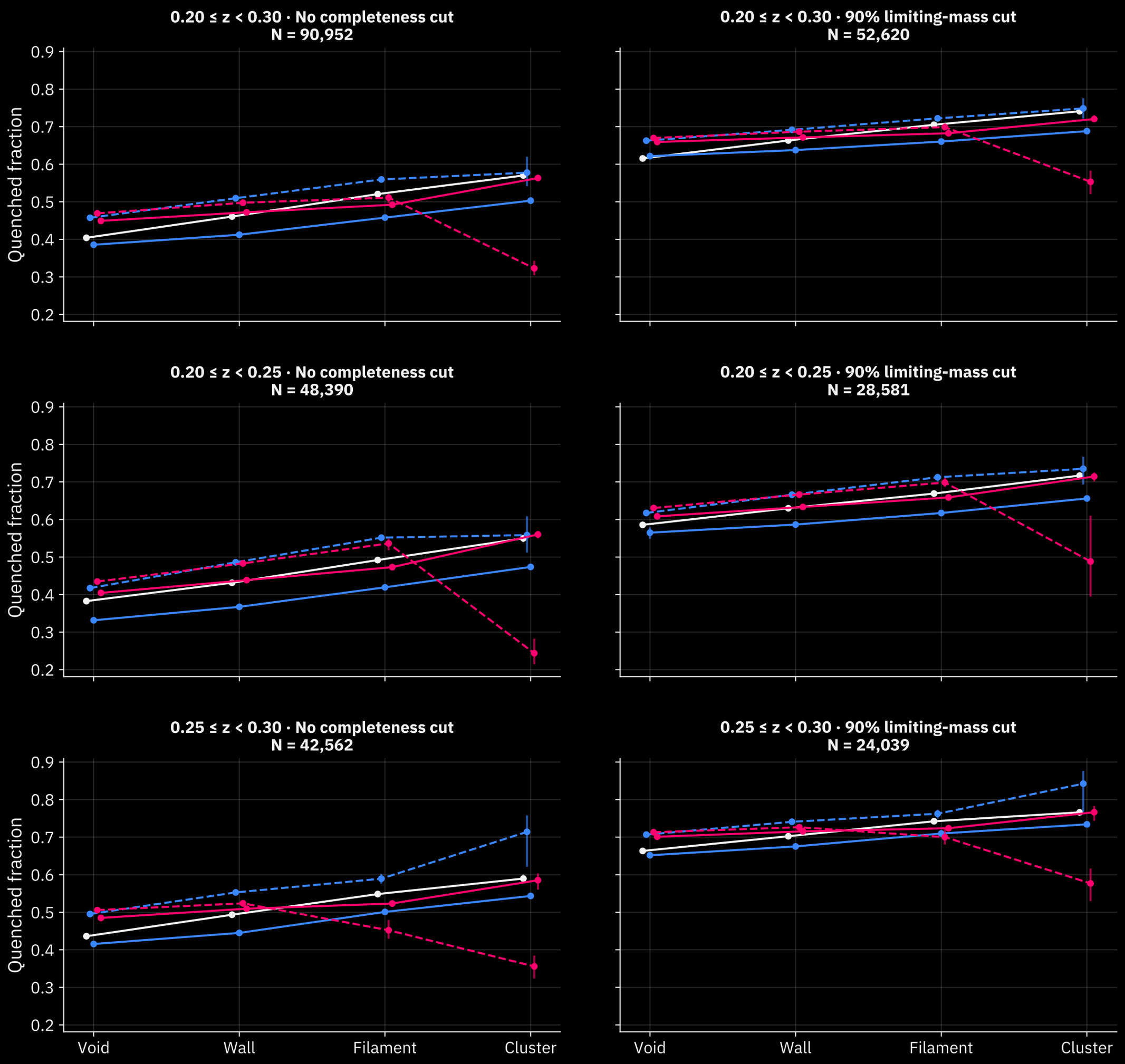
0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039



Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

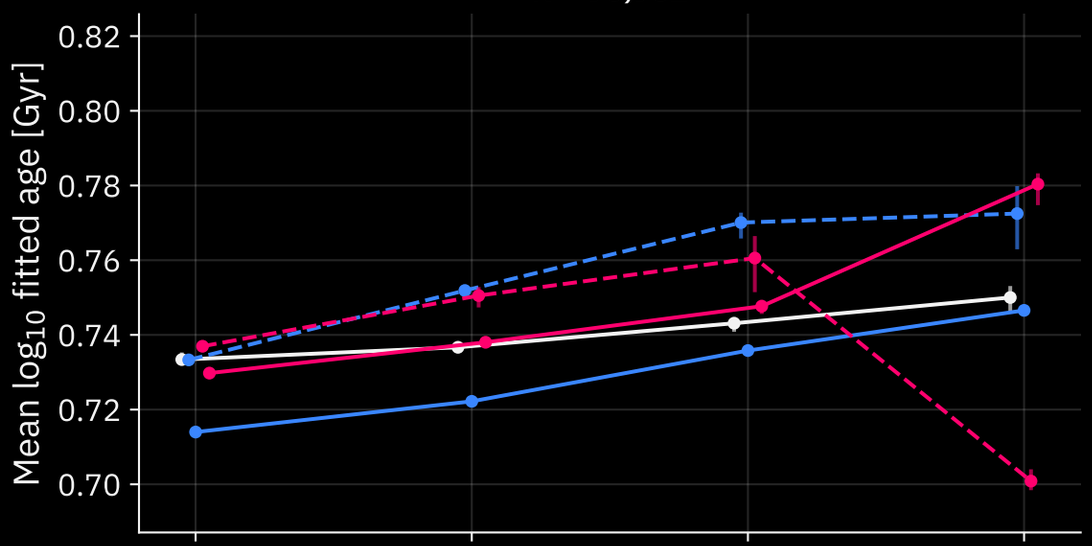
Actual TNG GAT versus Abacus P12-A · quenched
soft_raw · Frozen training-feature normalization



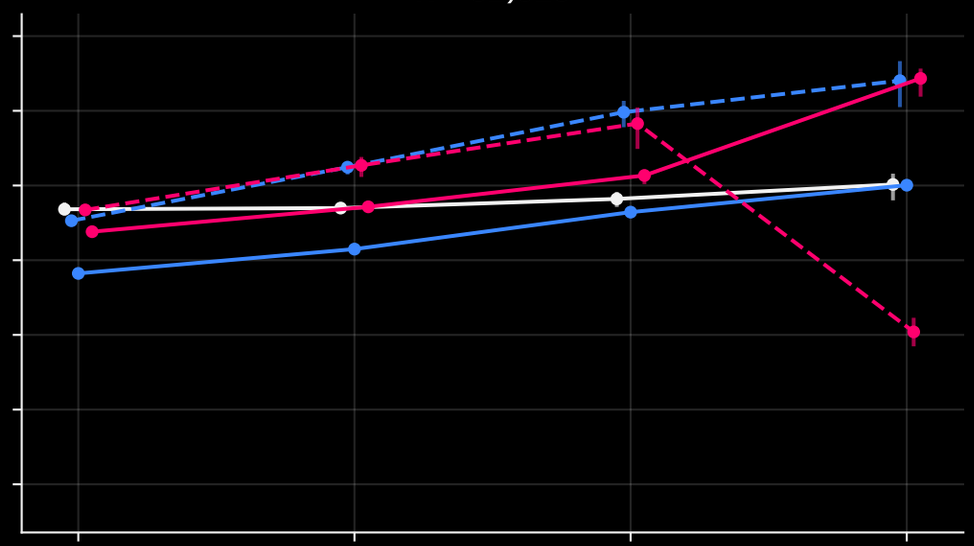
One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · age
soft_raw · Frozen training-feature normalization

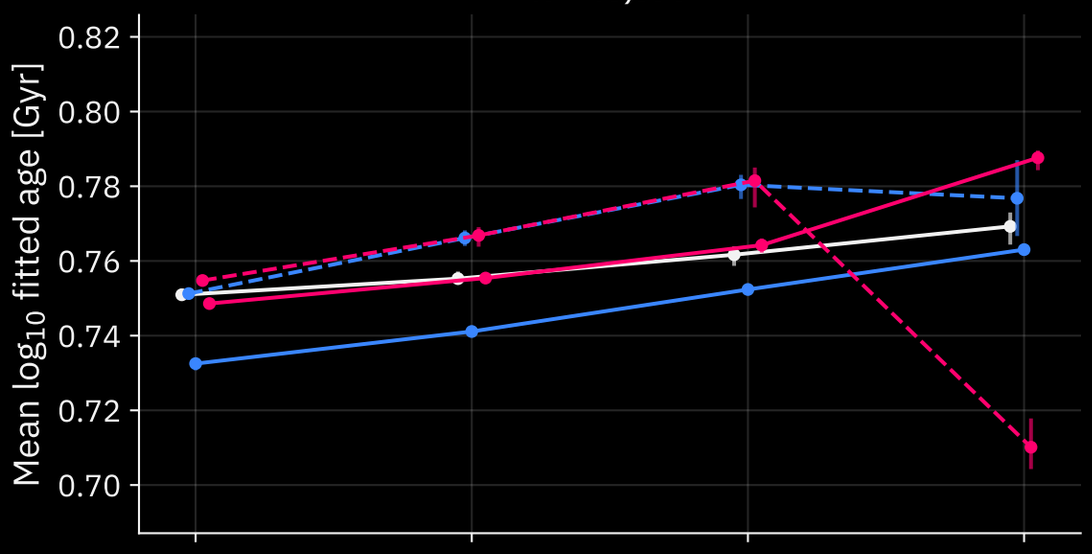
0.20 ≤ z < 0.30 · No completeness cut
N = 90,952



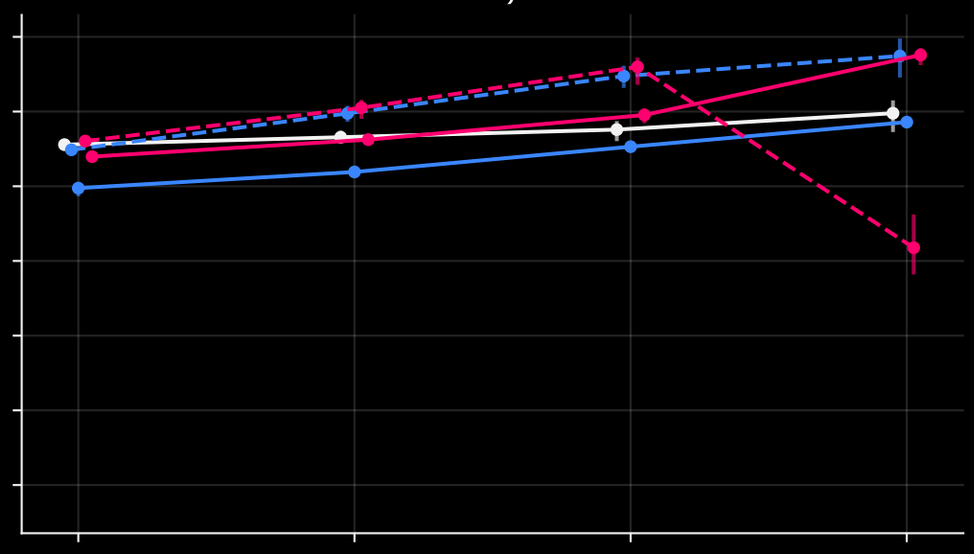
0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620



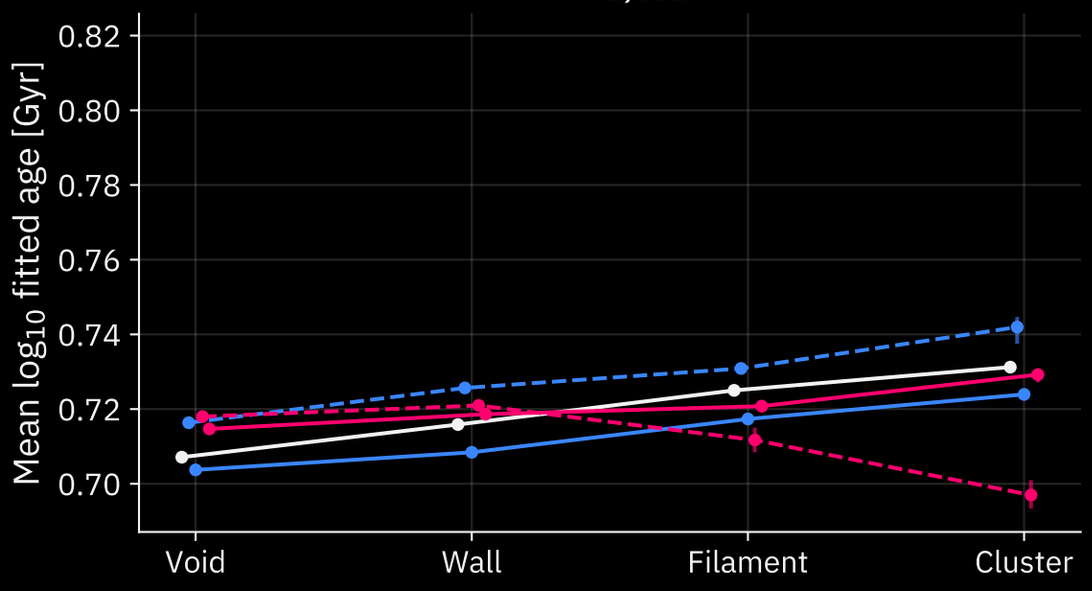
0.20 ≤ z < 0.25 · No completeness cut
N = 48,390



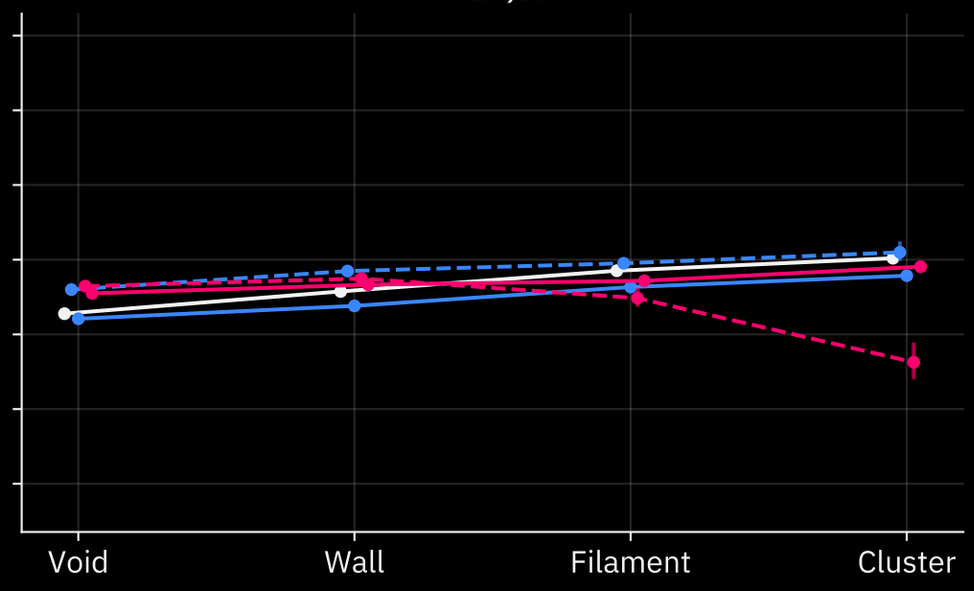
0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581



0.25 ≤ z < 0.30 · No completeness cut
N = 42,562



0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039

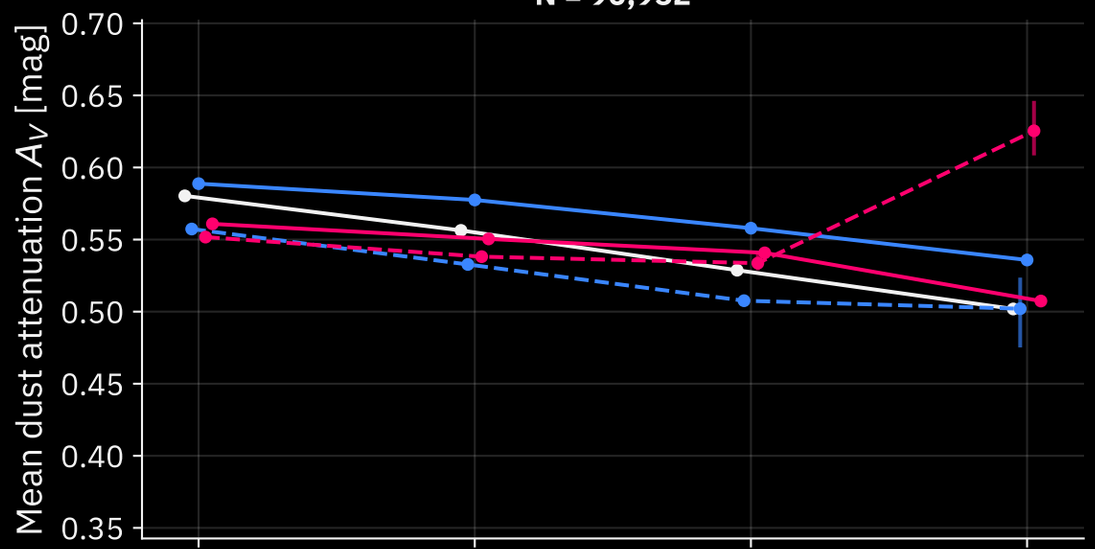


Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

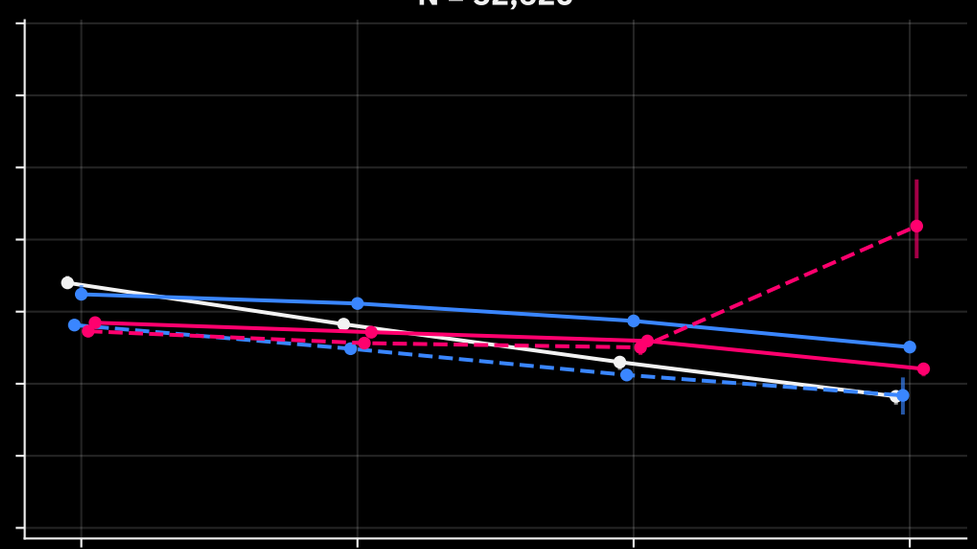
One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · av
soft_raw · Frozen training-feature normalization

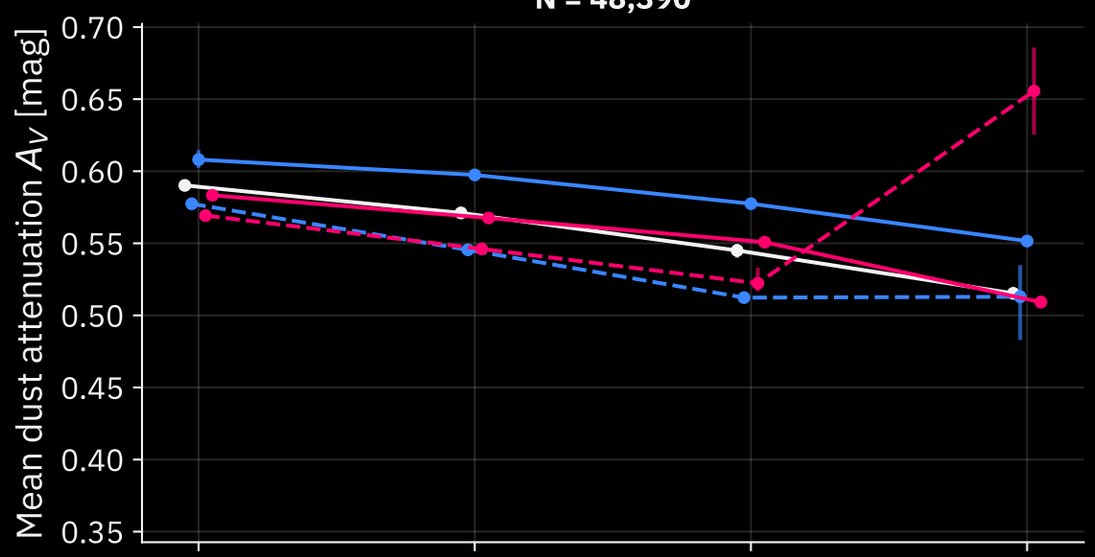
0.20 ≤ z < 0.30 · No completeness cut
N = 90,952



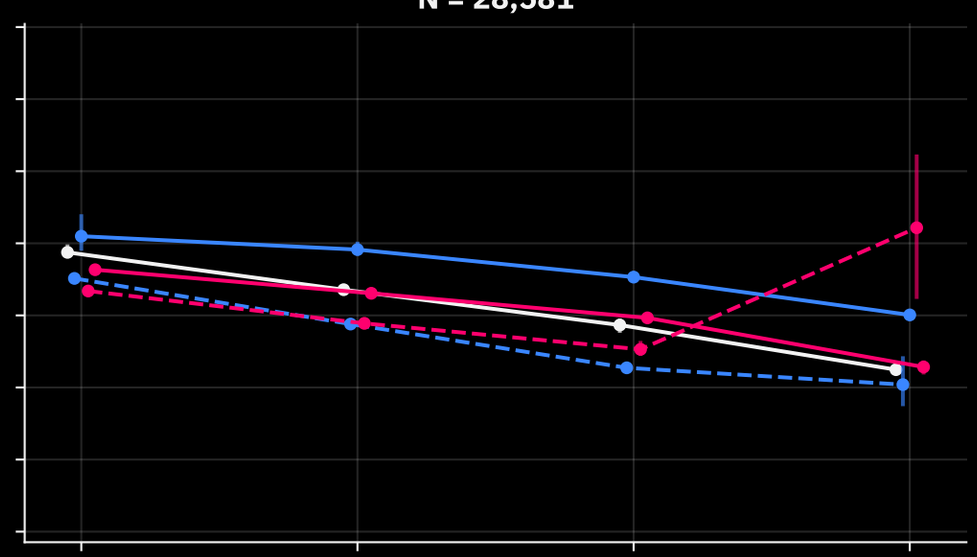
0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620



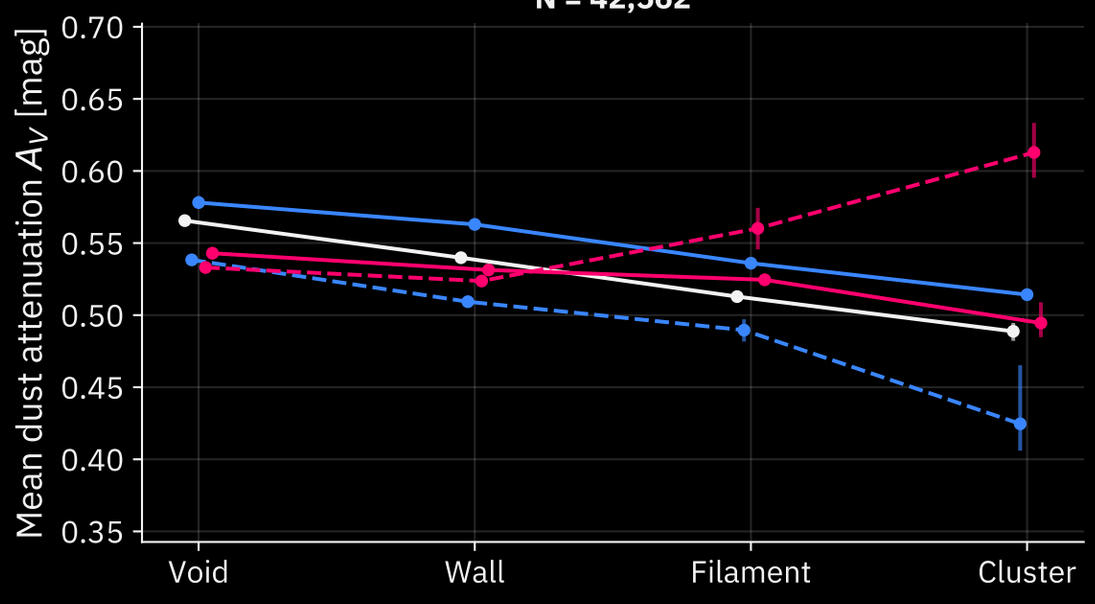
0.20 ≤ z < 0.25 · No completeness cut
N = 48,390



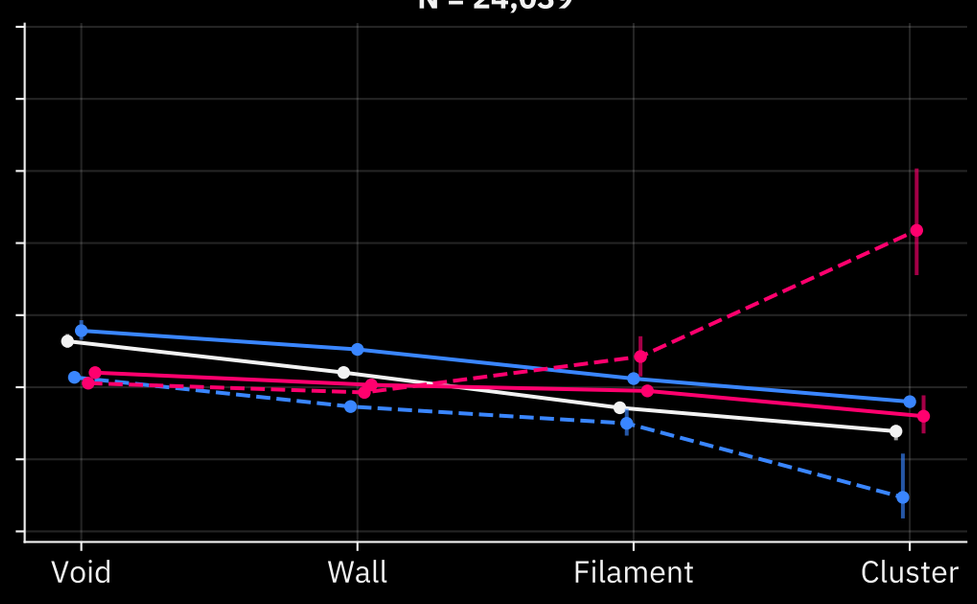
0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581



0.25 ≤ z < 0.30 · No completeness cut
N = 42,562



0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039



Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · fagn
soft_raw · Frozen training-feature normalization

0.20 ≤ z < 0.30 · No completeness cut
N = 90,952

0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620

0.20 ≤ z < 0.25 · No completeness cut
N = 48,390

0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581

0.25 ≤ z < 0.30 · No completeness cut
N = 42,562

0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039

Mean fitted AGN fraction

0.25
0.24
0.23
0.22
0.21
0.20
0.19

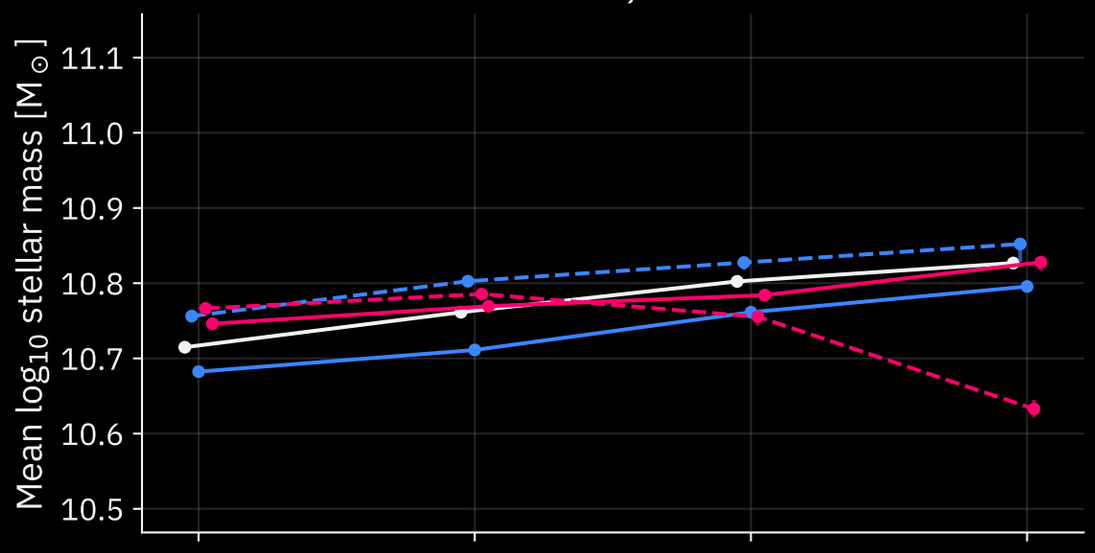
Void
Wall
Filament
Cluster

Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

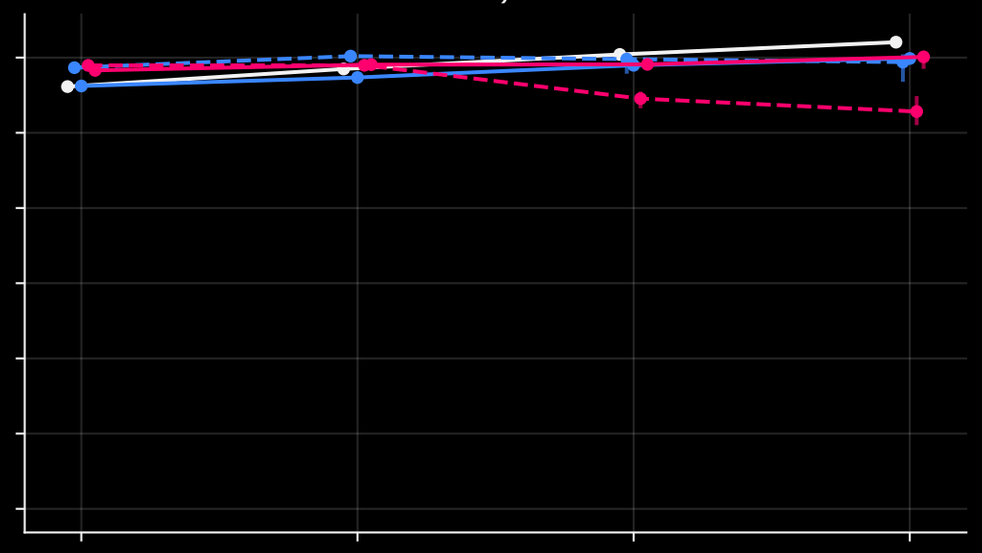
One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · mass
soft_controlled · Frozen training-feature normalization

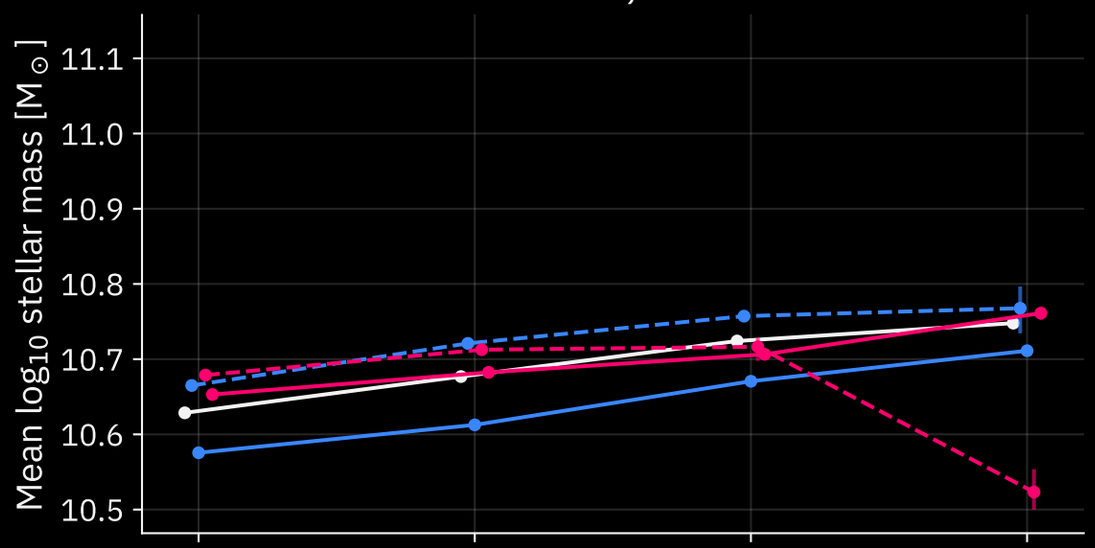
0.20 ≤ z < 0.30 · No completeness cut
N = 71,876



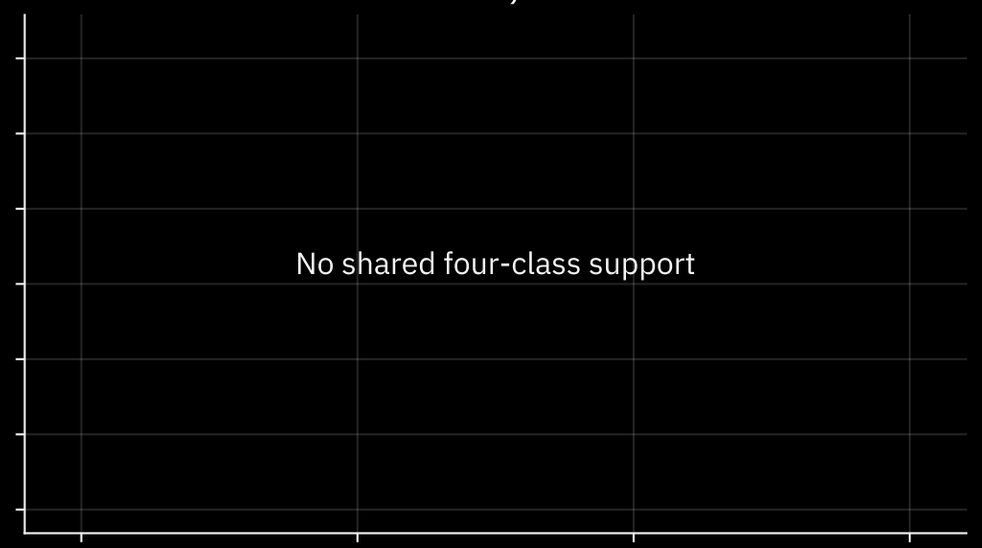
0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 7,902



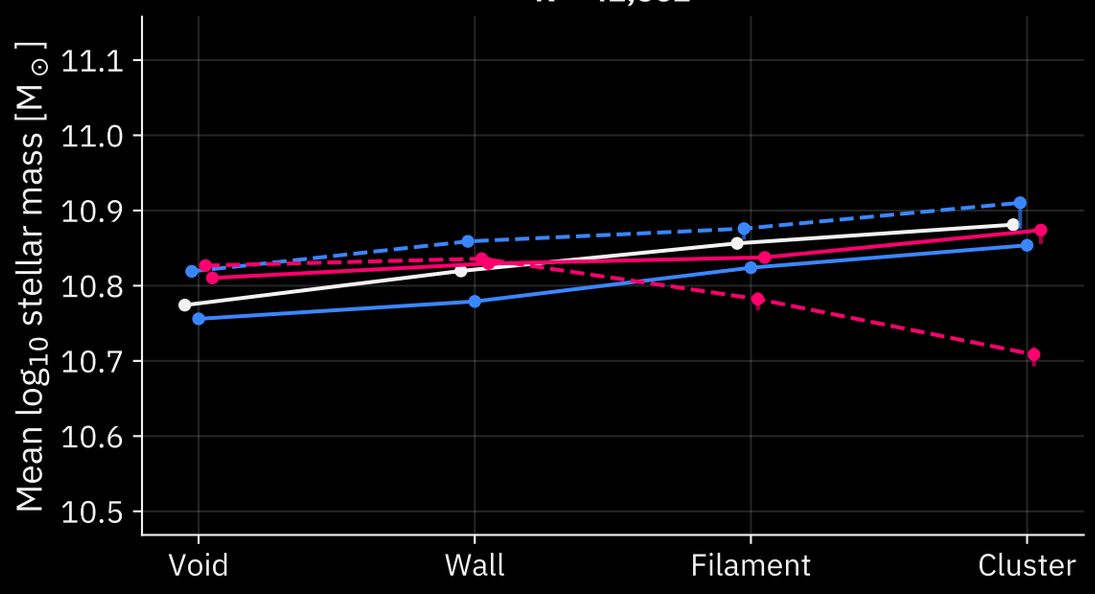
0.20 ≤ z < 0.25 · No completeness cut
N = 29,314



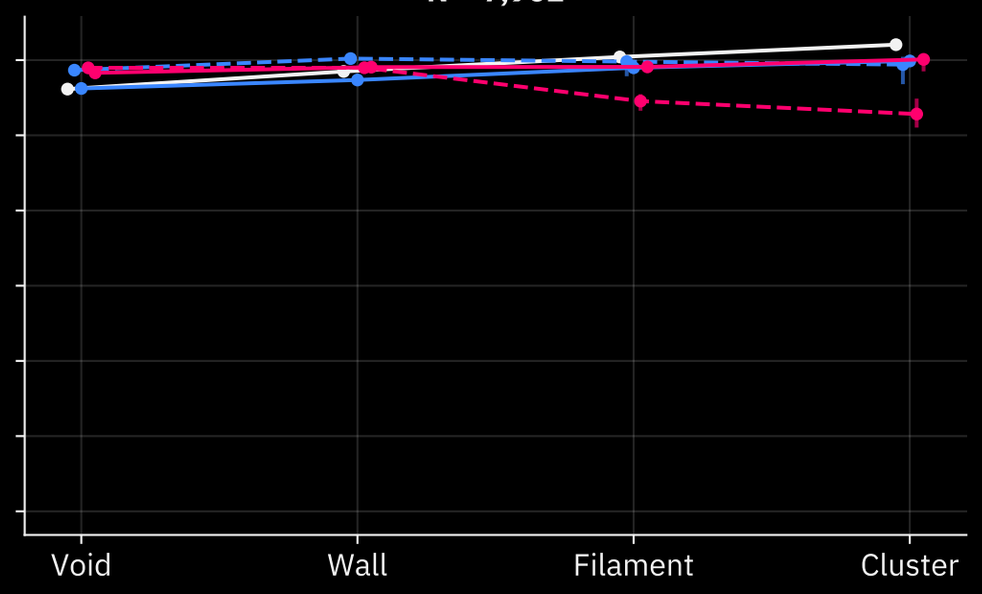
0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581



0.25 ≤ z < 0.30 · No completeness cut
N = 42,562



0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 7,902



Abacus P12-A · R7
GAT · R2 · normal
GAT · R2 · thinned
GAT · R7 · normal
GAT · R7 · thinned

One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · sfr
soft_controlled · Frozen training-feature normalization

0.20 ≤ z < 0.30 · No completeness cut
N = 90,952

No shared four-class support

0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620

No shared four-class support

0.20 ≤ z < 0.25 · No completeness cut
N = 48,390

No shared four-class support

0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581

No shared four-class support

0.25 ≤ z < 0.30 · No completeness cut
N = 42,562

No shared four-class support

0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039

No shared four-class support

Mean $\log_{10}$ SFR [$M_\odot$ yr^{-1}]

1.0
0.8
0.6
0.4
0.2
0.0

Void
Wall
Filament
Cluster

Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · ssfr
soft_controlled · Frozen training-feature normalization

0.20 ≤ z < 0.30 · No completeness cut
N = 90,952

No shared four-class support

0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620

No shared four-class support

0.20 ≤ z < 0.25 · No completeness cut
N = 48,390

No shared four-class support

0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581

No shared four-class support

0.25 ≤ z < 0.30 · No completeness cut
N = 42,562

No shared four-class support

0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039

No shared four-class support

Mean $\log_{10}$ sSFR [yr^{-1}]

1.0
0.8
0.6
0.4
0.2
0.0

Void · Wall · Filament · Cluster

Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

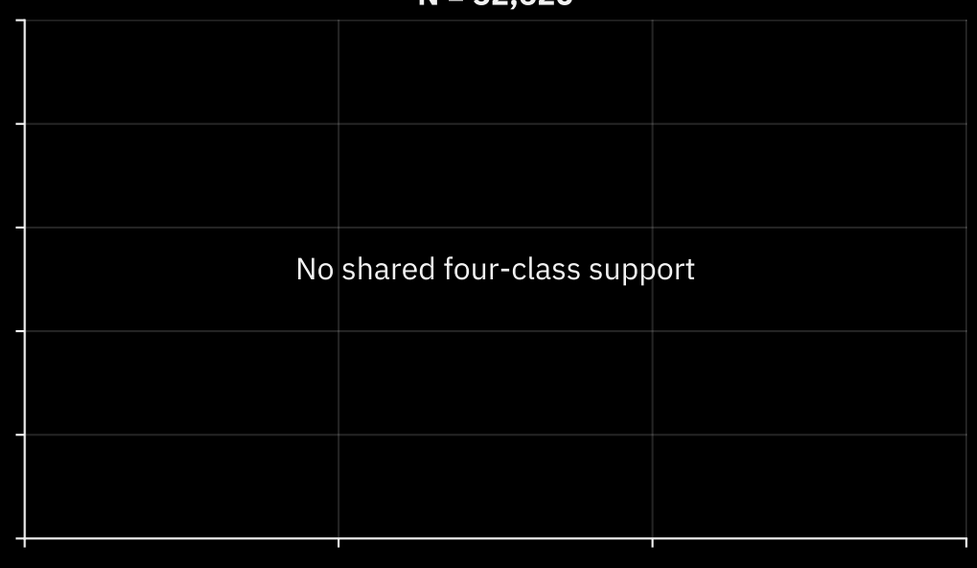
One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · quenched
soft_controlled · Frozen training-feature normalization

0.20 ≤ z < 0.30 · No completeness cut
N = 90,952



0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620



0.20 ≤ z < 0.25 · No completeness cut
N = 48,390



0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581



0.25 ≤ z < 0.30 · No completeness cut
N = 42,562



0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039





One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · age
soft_controlled · Frozen training-feature normalization

0.20 ≤ z < 0.30 · No completeness cut
N = 90,952

No shared four-class support

0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620

No shared four-class support

0.20 ≤ z < 0.25 · No completeness cut
N = 48,390

No shared four-class support

0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581

No shared four-class support

0.25 ≤ z < 0.30 · No completeness cut
N = 42,562

No shared four-class support

0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039

No shared four-class support

Mean $\log_{10}$ fitted age [Gyr]

1.0
0.8
0.6
0.4
0.2
0.0

Void · Wall · Filament · Cluster

Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · av
soft_controlled · Frozen training-feature normalization

0.20 ≤ z < 0.30 · No completeness cut
N = 90,952

No shared four-class support

0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620

No shared four-class support

0.20 ≤ z < 0.25 · No completeness cut
N = 48,390

No shared four-class support

0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581

No shared four-class support

0.25 ≤ z < 0.30 · No completeness cut
N = 42,562

No shared four-class support

0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039

No shared four-class support

Mean dust attenuation A_V [mag]

1.0
0.8
0.6
0.4
0.2
0.0

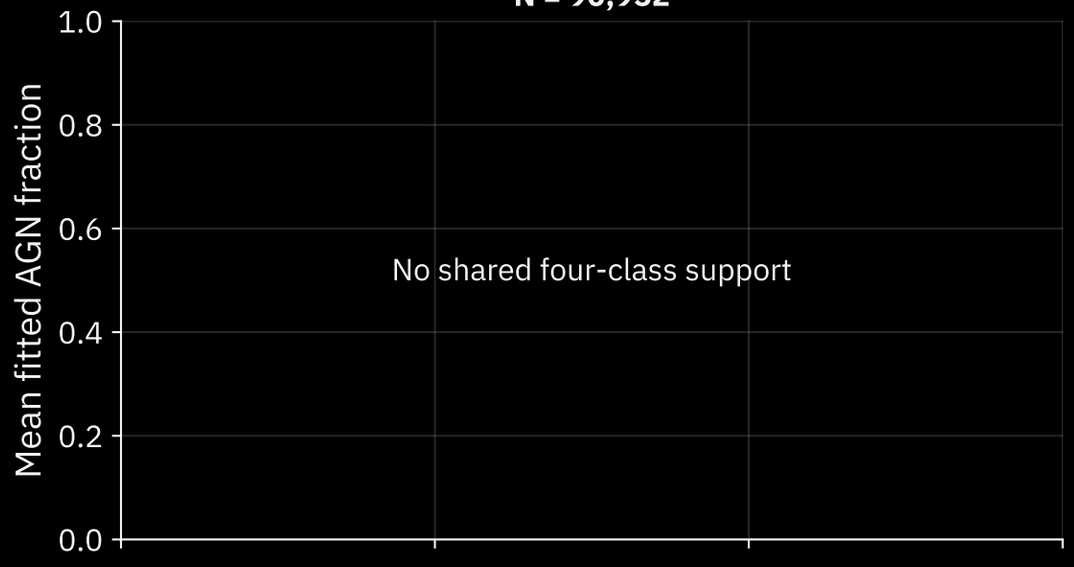
Void · Wall · Filament · Cluster

Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · fagn
soft_controlled · Frozen training-feature normalization

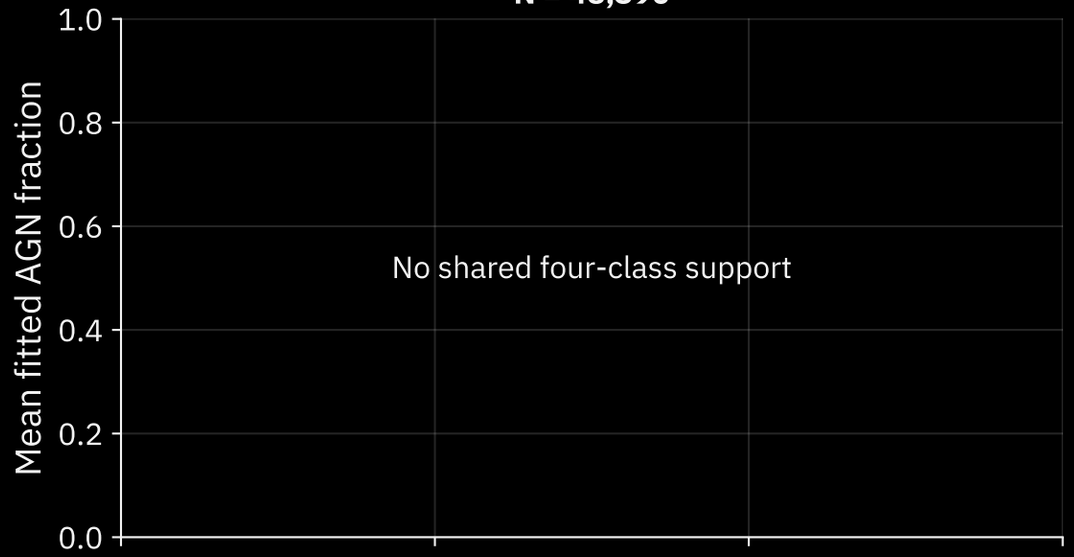
0.20 ≤ z < 0.30 · No completeness cut
N = 90,952



0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620



0.20 ≤ z < 0.25 · No completeness cut
N = 48,390



0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581



0.25 ≤ z < 0.30 · No completeness cut
N = 42,562



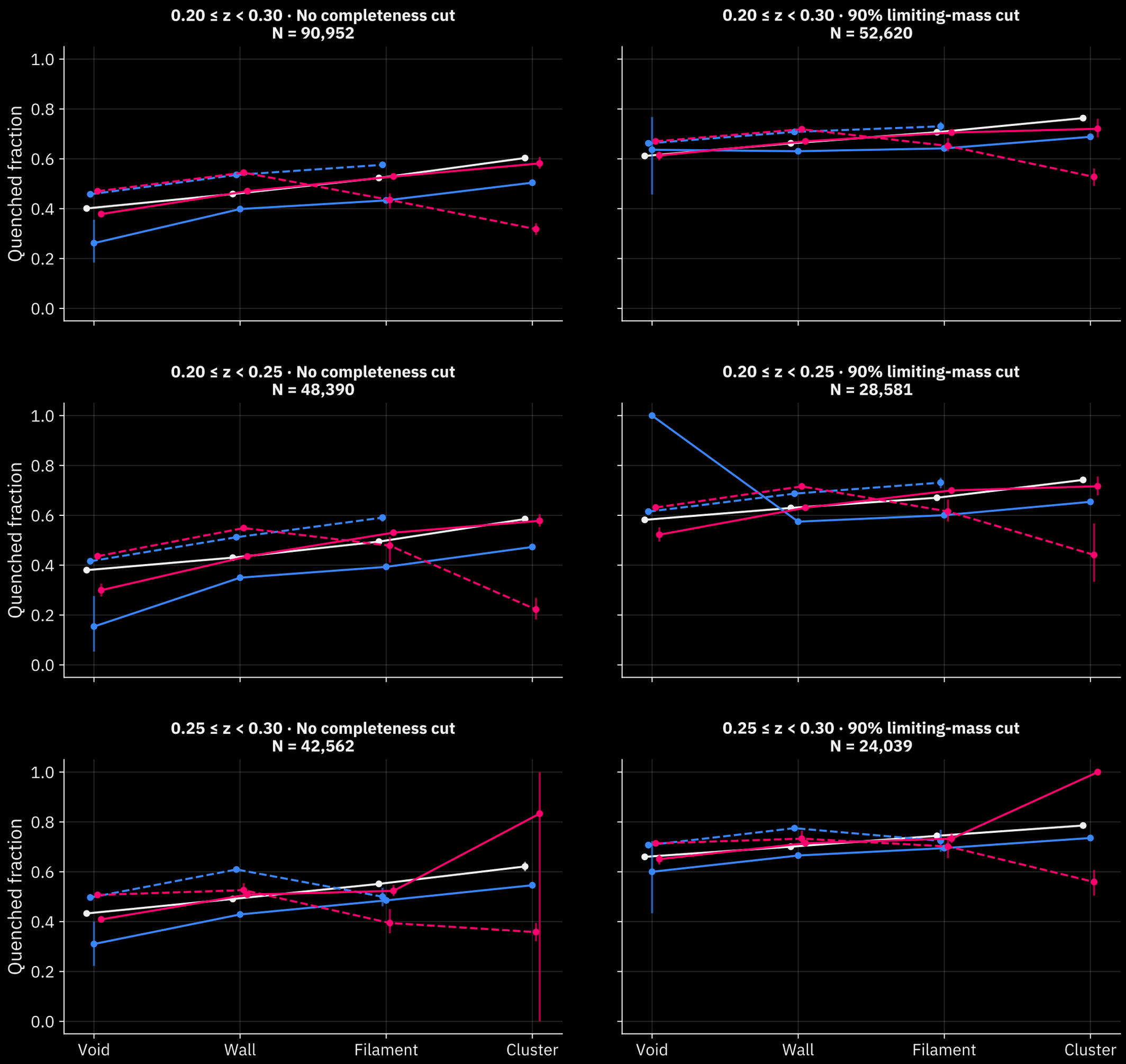
0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039



Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · quenched
hard_raw · Frozen training-feature normalization



One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.

Actual TNG GAT versus Abacus P12-A · quenched
hard_controlled · Frozen training-feature normalization

0.20 ≤ z < 0.30 · No completeness cut
N = 90,952



0.20 ≤ z < 0.30 · 90% limiting-mass cut
N = 52,620



0.20 ≤ z < 0.25 · No completeness cut
N = 48,390



0.20 ≤ z < 0.25 · 90% limiting-mass cut
N = 28,581



0.25 ≤ z < 0.30 · No completeness cut
N = 42,562



0.25 ≤ z < 0.30 · 90% limiting-mass cut
N = 24,039



Abacus P12-A · R7
GAT · R2 · thinned
GAT · R7 · thinned
GAT · R2 · normal
GAT · R7 · normal

One training seed per GAT; 128 sky-block resamples (16–84%). Class scores are not calibrated observational truth.